

Principles of Distributed Systems

第5周：分布式系统中的通信与命名
Week 5: Communication and Naming in Distributed Systems

By: Fahad Sabah

计算机学院,
北京工业大学, 中国
College of Computer Science,
Beijing University of Technology, China

fahad.sabah@bjut.edu.cn



Recap of Week3

第三周回顾

1. Processes & Threads Basics

2. Multithreading in Distributed Systems

3. Virtualization (VMs, Containers)

4. Multi-threaded client-server + virtualization setup

1. 流程与线程基础

2. 分布式系统中的多线程

3. 虚拟化（虚拟机、容器）

4. 多线程客户端-服务器+虚拟化设置

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1. Communication in Distributed Systems

2. Naming in Distributed Systems

3. Midterm Project Discussion

4. Moving Towards Technical Research Contribution (Final Project)

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1. 分布式系统中的通信

2. 分布式系统中的命名

3. 期中项目讨论

4. 迈向技术研究贡献（毕业项目）

分布式系统中的通信

Communication in Distributed Systems

Why Communication Matters in Distributed Systems

- ❑ Nodes must exchange data, coordinate actions, maintain consistency

❑ Core functions:

- ❑ Inter-node coordination (share state, synchronize operations)
- ❑ Data consistency (integrity across replicas)
- ❑ Fault tolerance (error detection, retries, recovery)
- ❑ Scalability (add nodes without performance loss)

为什么通信在分布式系统中很重要

- ❑ 节点必须交换数据、协调操作、保持一致性

核心职能：

- ❑ 节点间协调（共享状态，同步操作）
- ❑ 数据一致性（副本间的完整性）
- ❑ 容错（错误检测、重试、恢复）
- ❑ 可扩展性（添加节点而不影响性能）

分布式系统中的通信

Communication in Distributed Systems

Communication Fundamentals

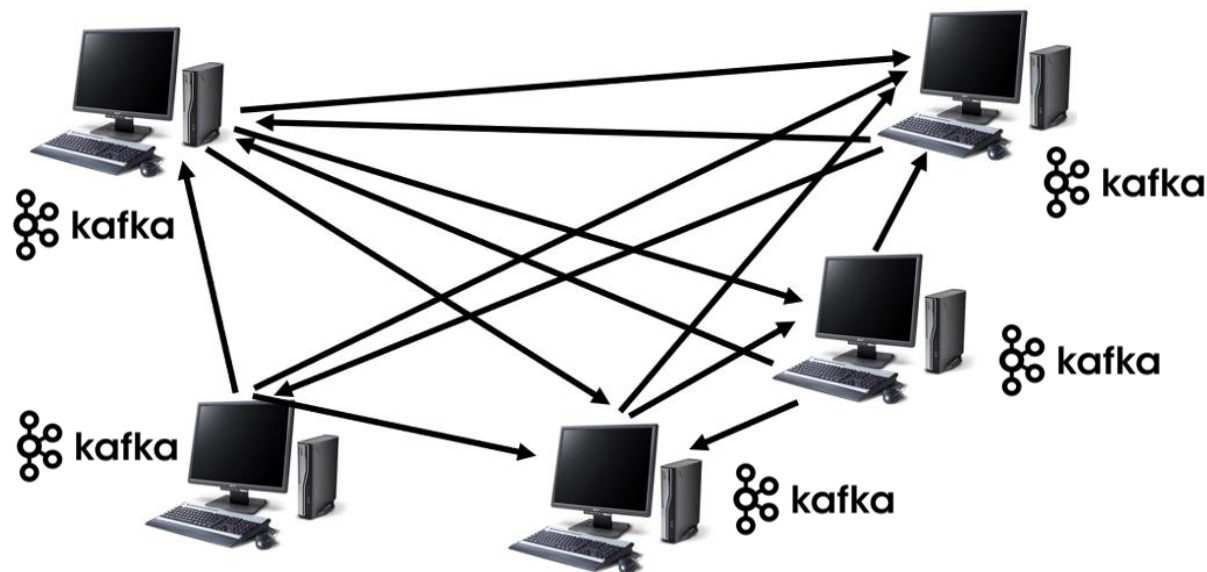
传播基础

RPC and Message-Oriented Communication

RPC与面向消息通信

Session 1: Introduction & Remote Procedure Call (RPC)

第一场：介绍与远程程序通话（RPC）



Week 5: Communication and Naming in Distributed Systems

第5周：分布式系统中的通信与命名

分布式系统中的通信

Communication in Distributed Systems

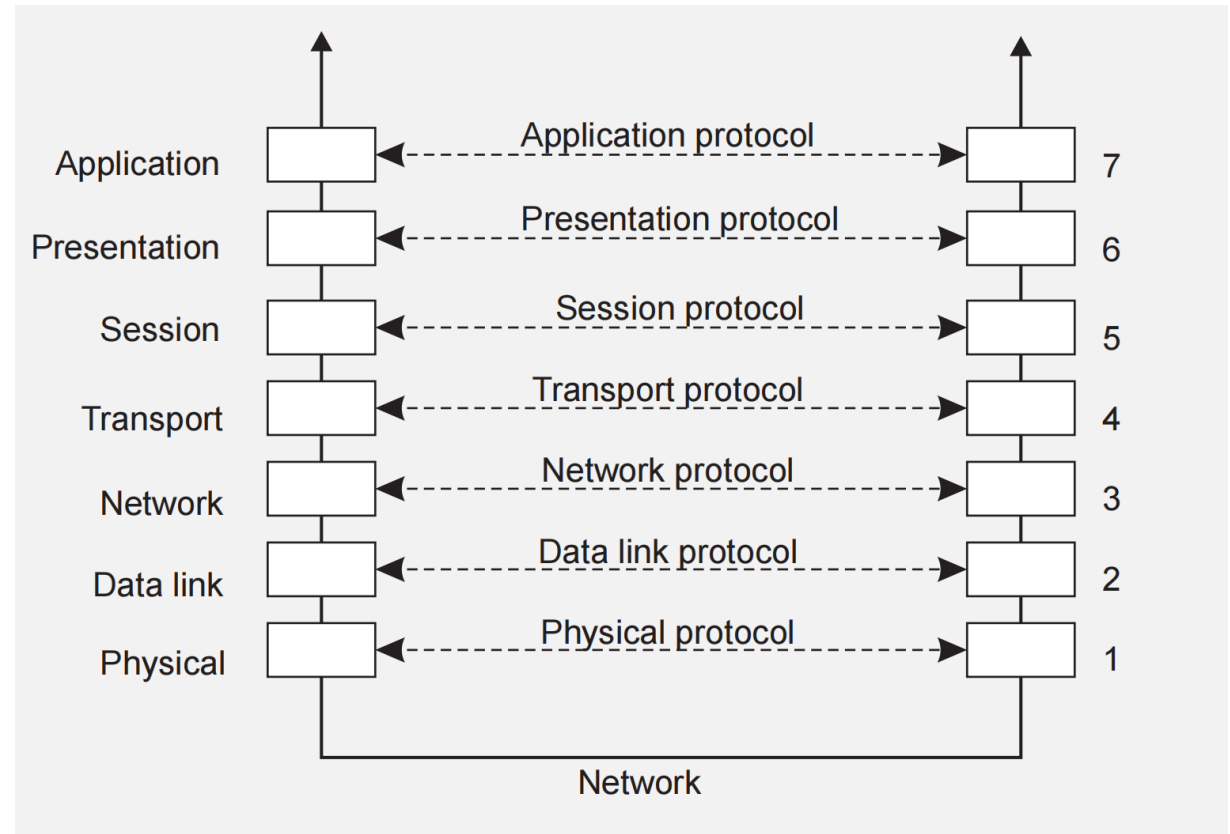
Communication Protocols Overview

- ❑ Layered approach:
 - ❑ Low-level layers
 - ❑ Transport layer
 - ❑ Application layer
 - ❑ Middleware layer

Drawbacks

- ❑ Focus on message-passing only
- ❑ Often unneeded or unwanted functionality
- ❑ Violates access transparency

Basic networking model



分布式系统中的通信

Communication in Distributed Systems

Low-level layer

Physical layer: contains the specification and implementation of bits, and their transmission between sender and receiver

Data link layer: prescribes the transmission of a series of bits into a frame to allow for error and flow control

Network layer: describes how packets in a network of computers are to be routed.

低层

物理层：包含比特的规格和实现，以及发送端和接收端之间的传输

数据链路层：规定将一系列比特传输到帧中，以实现误差和流量控制

网络层：描述计算机网络中数据包的路由方式。

分布式系统中的通信

Communication in Distributed Systems

Transport Layer

The transport layer provides the actual communication facilities for most distributed systems.

Standard Internet protocols

- ❑ TCP: connection-oriented, reliable, stream-oriented communication
- ❑ UDP: unreliable datagram communication

传输层

传输层为大多数分布式系统提供了实际的通信设施。

标准互联网协议

TCP: 面向连接、可靠、流向通信

UDP: 不可靠（尽力而为）数据报通信

分布式系统中的通信

Communication in Distributed Systems

Middleware Layer

Middleware is invented to provide common services and protocols that can be used by many applications

A rich set of communication protocols

- ❑ (Un)marshaling of data, necessary for integrated systems
- ❑ Naming protocols, to allow easy sharing of resources
- ❑ Security protocols for secure communication
- ❑ Scaling mechanisms, such as for replication and caching

中间件层

中间件的发明旨在提供通用的服务和协议，供多种不同应用使用。

丰富的通信协议集

- ❑ 数据的（非）分组，集成系统所必需
- ❑ 命名协议，以便资源共享
- ❑ 用于安全通信的安全协议
- ❑ 缩放机制，如复制和缓存

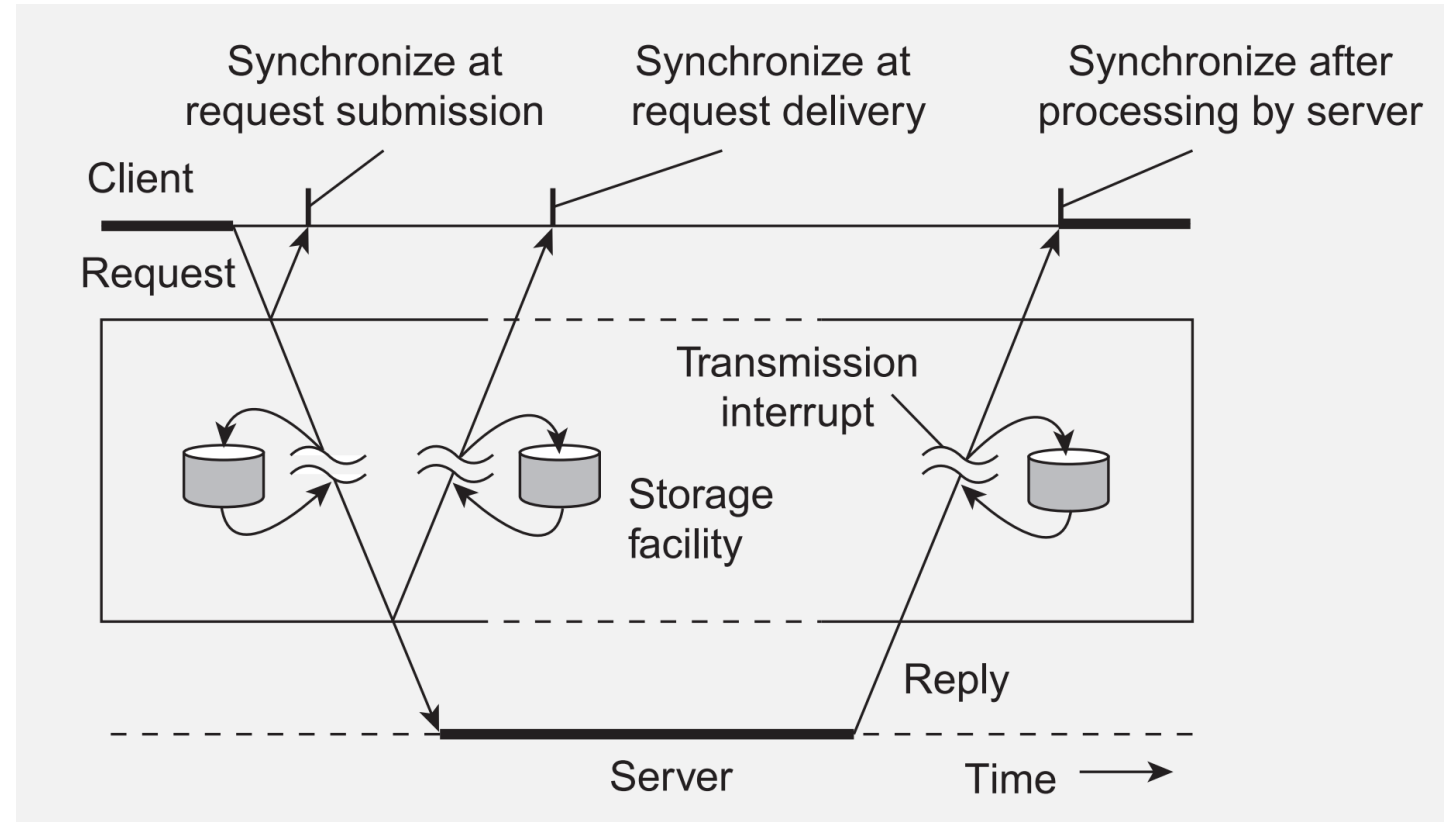
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Communication in Distributed Systems

Types of communication

Differentiate

- ☐ Transient versus persistent communication
- ☐ Asynchronous versus synchronous communication



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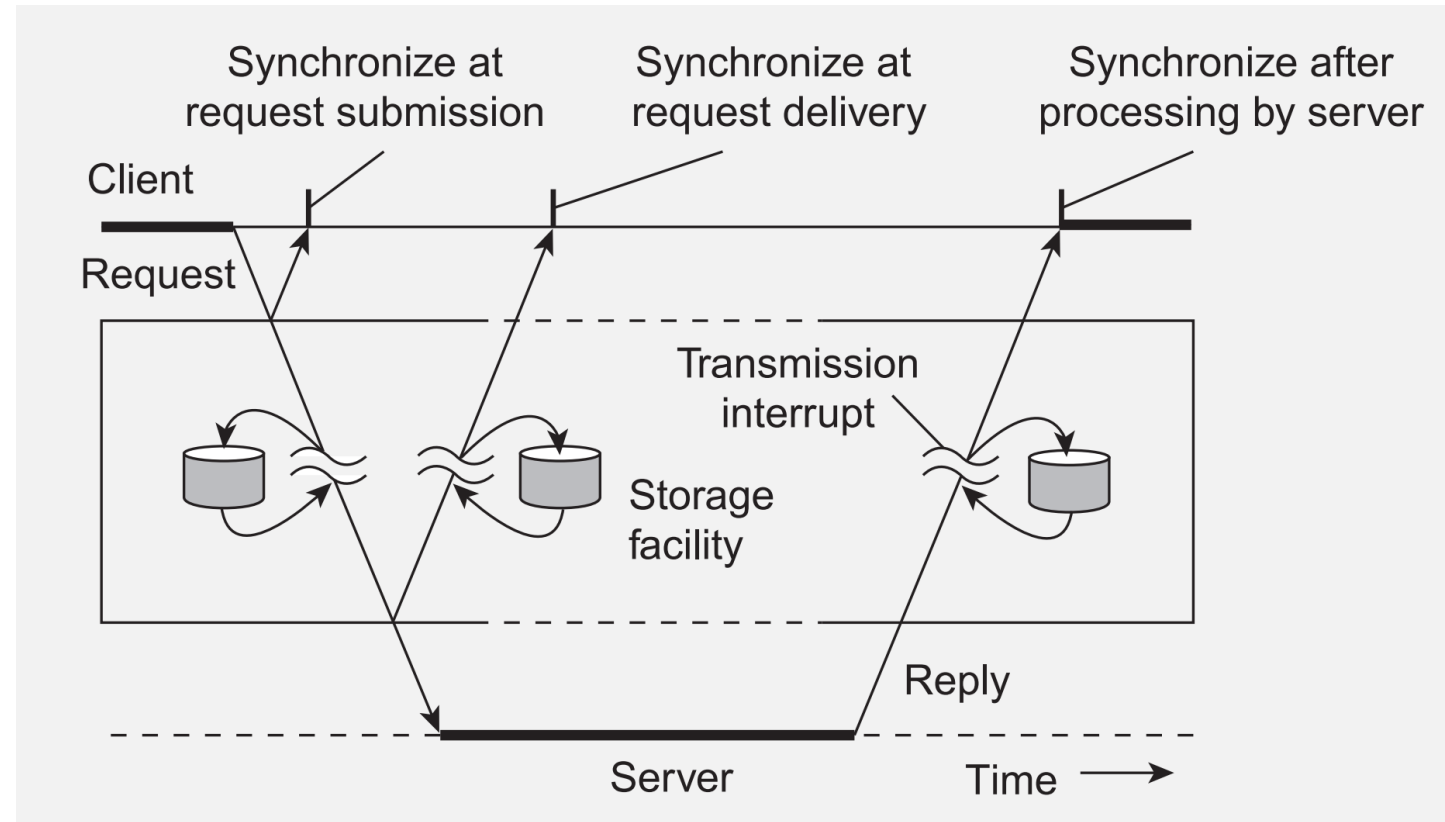
Communication in Distributed Systems

Types of communication

Transient versus persistent

Transient communication: Communication server discards message when it cannot be delivered at the next server, or at the receiver.

Persistent communication: A message is stored at a communication server as long as it takes to deliver it.



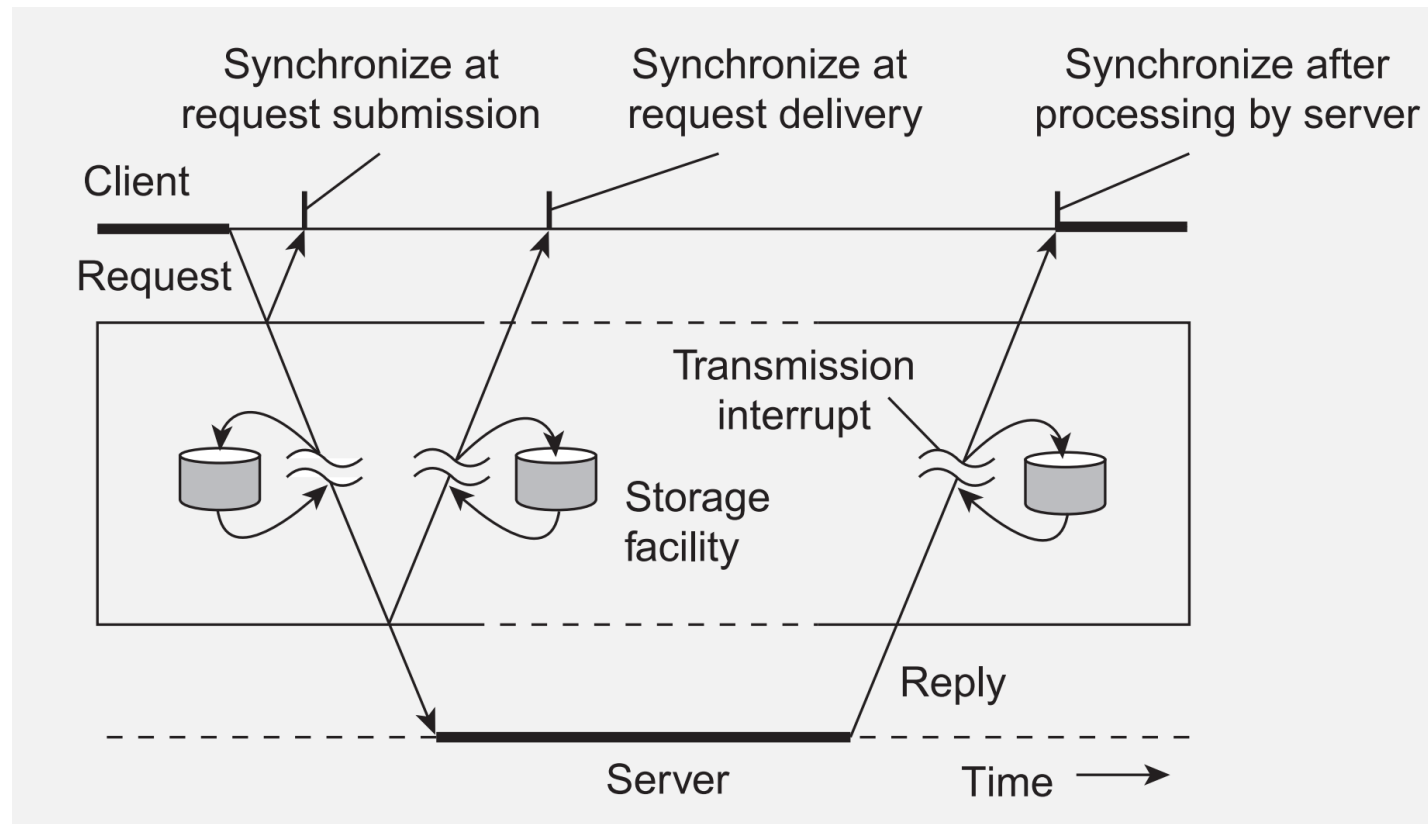
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Communication in Distributed Systems

Types of communication

Places for synchronization

- ☐ At request submission
- ☐ At request delivery
- ☐ After request processing

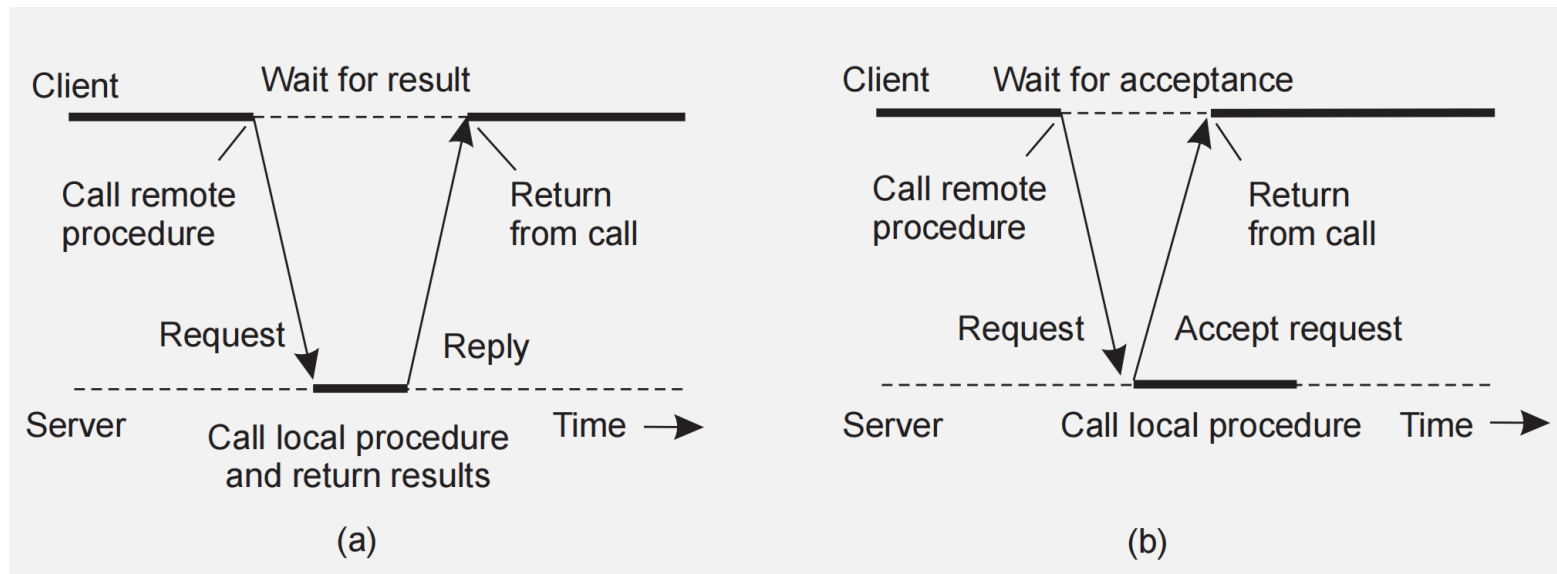


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Communication in Distributed Systems

Asynchronous RPCs

Try to get rid of the strict request-reply behavior, but let the client continue without waiting for an answer from the server.



分布式系统中的通信

Communication in Distributed Systems

Client/Server

Client/Server computing is generally based on a model of transient synchronous communication:

- ❑ Client and server have to be active at time of communication.
- ❑ Client issues request and blocks until it receives reply
- ❑ Server essentially waits only for incoming requests, and subsequently processes them

Drawbacks synchronous communication

- ❑ Client cannot do any other work while waiting for reply
- ❑ Failures have to be handled immediately: the client is waiting

客户端/服务器

客户端/服务器计算通常基于瞬时同步通信模型：

- ❑ 客户端和服务端必须在通信时处于激活状态。
- ❑ 客户端发出请求并屏蔽，直到收到回复
- ❑ 服务器本质上只等待收到的请求，然后处理它们

缺点：同步通信

- ❑ 客户在等待回复期间无法进行其他工作
- ❑ 故障必须立即处理：客户端正在等待

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Communication in Distributed Systems

Messaging

Message-oriented middleware

Aims at high-level persistent asynchronous communication:

- ❑ Processes send each other messages, which are queued
- ❑ Sender need not wait for immediate reply, but can do other things
- ❑ Middleware often ensures fault tolerance

信息传递

面向消息中间件

旨在实现高层次的持久异步通信:

- ❑ 进程相互发送消息，这些消息被排队
- ❑ 发送方无需等待立即回复，可以做其他事情
- ❑ 中间件通常能保证容错

分布式系统中的通信

Communication in Distributed Systems

Key mechanisms for distributed systems:

- ❑ RPC (e.g., gRPC, Thrift) – remote functions as local calls
- ❑ Message passing (e.g., MQTT, AMQP) – structured messages
- ❑ Publish-subscribe (e.g., Kafka, Redis Pub/Sub) – decoupled via topics
- ❑ ORBs (e.g., CORBA) – distributed object interactions
- ❑ SOA protocols (e.g., SOAP) – service-to-service

分布式系统的关键机制：

- ❑ RPC（例如， gRPC、Thrift）—远程作为本地调用
- ❑ 消息传递（例如， MQTT、AMQP）—结构化消息
- ❑ 发布订阅（例如， Kafka， Redis Pub/Sub）—通过主题脱钩
- ❑ ORB（例如 CORBA）—分布式对象交互
- ❑ SOA协议（例如SOAP）—服务到服务

分布式系统中的通信

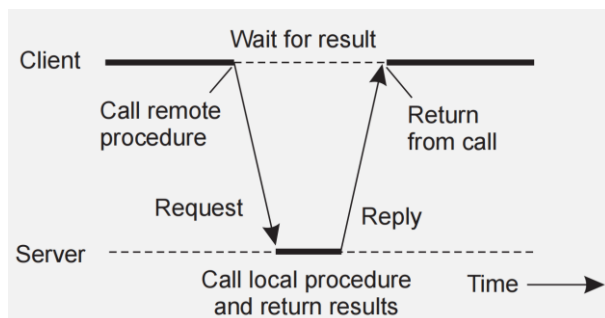
Communication in Distributed Systems

Remote Procedure Call (RPC) – Concept

Idea: Call a function on a remote machine as if it were local

How RPC Works – Step by Step

1. Client calls local stub
2. Stub marshals parameters into a message
3. Message sent via transport (e.g., TCP, HTTP/2)
4. Server stub unmarshals and calls actual procedure
5. Result marshaled and sent back
6. Client stub unmarshals and returns to caller



远程过程调用 (RPC) ——概念

想法：在远程机器上调用一个函数，就像它是本地设备一样

RPC的工作原理一步步

1. 客户端调用本地存根
2. 存根将参数编组成消息
3. 通过传输（例如 TCP，HTTP/2）发送的消息
4. 服务器存根解封并调用实际过程
5. 结果经过管理并送回
6. 客户存根取消管理并返回来电者

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Popular RPC Frameworks

1. gRPC – uses Protocol Buffers, HTTP/2, supports streaming
2. Apache Thrift – cross-language, multiple transports
3. JSON-RPC / XML-RPC – simple, text-based
4. Java RMI – Java-to-Java only

流行RPC框架

1. gRPC – 使用协议缓冲区, HTTP/2, 支持流式传输
2. Apache 节俭——跨语言, 多传输
3. JSON-RPC / XML-RPC – 简单, 基于文本
4. Java RMI – 仅 Java 对 Java

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Code Example – gRPC (Python)

```
import grpc
import myservice_pb2
import myservice_pb2_grpc

# Create channel and stub
channel = grpc.insecure_channel('localhost:50051')
stub = myservice_pb2_grpc.MyServiceStub(channel)

# Make remote call
response =
stub.MyMethod(myservice_pb2.MyRequest(param="Hello"))
print("Server Response:", response.message)
```

代码示例 – gRPC (Python)

注：该小作品摘要了所有网络细节

Note: The stub abstracts all network details

Communication between caller & callee can be hidden by using procedure-call mechanism.

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RPC – Advantages & Challenges

Advantages:

- ☐ Simple programming model (looks like local call)
- ☐ Efficient (binary protocols, multiplexing)
- ☐ Strong typing with IDL (e.g., Protobuf)

Challenges:

- ☐ Partial failures (network, server down)
- ☐ Location transparency can mask latency
- ☐ Tight coupling in time (blocking call by default)

RPC—优势与挑战

优势:

- ☐ 简单编程模型（看起来像本地调用）
- ☐ 高效（二进制协议，复用）
- ☐ 带 IDL 的强类型（例如 Protobuf）

挑战:

- ☐ 部分故障（网络、服务器宕机）
- ☐ 位置透明度可以掩盖延迟
- ☐ 时间紧耦合（默认阻断呼叫）

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Communication in Distributed Systems

Why Not Always RPC?

RPC works well for request-response, but has limitations:

- ❑ Synchronous blocking may hurt scalability
- ❑ Tight coupling (client & server both must be up)
- ❑ Buffering / load leveling not built-in

Solution: Message-oriented communication

为什么不总是用RPC?

RPC 在请求-响应方面表现良好，但存在局限性：

- ❑ 同步分流可能会影响可扩展性
- ❑ 紧耦合（客户端和服务端必须同时在线）
- ❑ 缓冲/负载平稳并非内置功能

解决方案：面向消息的通信

分布式系统中的通信

Communication in Distributed Systems

Message-Oriented Communication – Core Ideas

Message passing: Nodes exchange discrete messages via channels

Two styles:

- ❑ Point-to-point (queue) – one sender, one receiver
- ❑ Publish-subscribe (topic) – one sender, many receivers

Synchronous (sender waits for ack) vs. Asynchronous (fire-and-forget)

以信息为导向的沟通——核心理念

消息传递：节点通过通道交换离散消息

两种风格：

- ❑ 点对点（队列）一个发送者，一个接收者
- ❑ 发布订阅（主题）一个发送者，多个接收者

同步（发送方等待ack）与异步（发射后遗忘）

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Communication in Distributed Systems

Point-to-Point Message Passing

Example: Message queue (RabbitMQ, ActiveMQ)

Properties:

- ☐ Message removed after being consumed
- ☐ Load balancing across multiple consumers
- ☐ Reliable delivery (persistence, acknowledgments)

Use cases: task queues, order processing

点对点消息传递

示例：消息队列（RabbitMQ, ActiveMQ）

房产：

- ☐ 消息被消费后删除
- ☐ 跨多个消费者的负载均衡
- ☐ 可靠传递（持久性、确认）

用例：任务队列、订单处理

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Communication in Distributed Systems

Publish-Subscribe (Pub/Sub)

- ❑ Decoupling in time, space, and synchronization
- ❑ Publishers send messages to topics (not specific receivers)
- ❑ Subscribers receive all messages on topics they subscribe to

发布-订阅（出版/订阅）

- ❑ 时间、空间与同步上的解耦
- ❑ 出版商会向主题发送消息（不是特定的接收者）
- ❑ 订阅者会收到他们订阅主题的所有消息

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Communication in Distributed Systems

Comparing Communication Mechanisms

比较通信机制

Feature	RPC	Message Queue	Pub/Sub
Coupling	Tight (call-response)	Loose (temporal)	Very loose
Buffering	No (unless client retries)	Yes (queue)	Yes (if broker persistent)
Scalability	Moderate (connection per call)	High (async consumers)	Very high
Typical use	Microservice internal calls	Task distribution	Event broadcasting

分布式系统中的通信

Communication in Distributed Systems

Security in Communication – Overview

- ❑ Confidentiality – prevent eavesdropping (AES, RSA)
- ❑ Integrity – detect tampering (SHA-256, HMAC)
- ❑ Authentication – verify identity (X.509 certificates, JWT)
- ❑ Non-repudiation – sender cannot deny action (digital signatures)
- ❑ Authorization – control access (ACL, RBAC)

通信安全—概述

- ❑ 保密性—防止窃听 (AES, RSA)
- ❑ 完整性—检测篡改 (SHA-256, HMAC)
- ❑ 认证—验证身份 (X.509证书, JWT)
- ❑ 不可否认—发送方不能拒绝采取行动 (数字签名)
- ❑ 授权—控制访问 (ACL, RBAC)

分布式系统中的通信

Communication in Distributed Systems

Securing RPC and Message-Oriented Systems

For RPC:

- ☐ Use TLS (server/client authentication)
- ☐ Token-based auth (OAuth2, JWT)

For message brokers (Kafka, RabbitMQ):

- ☐ SSL/TLS for transport encryption
- ☐ SASL for authentication
- ☐ ACLs for topic/queue permissions

保护RPC和面向消息系统的安全

对于RPC（例如gRPC）：

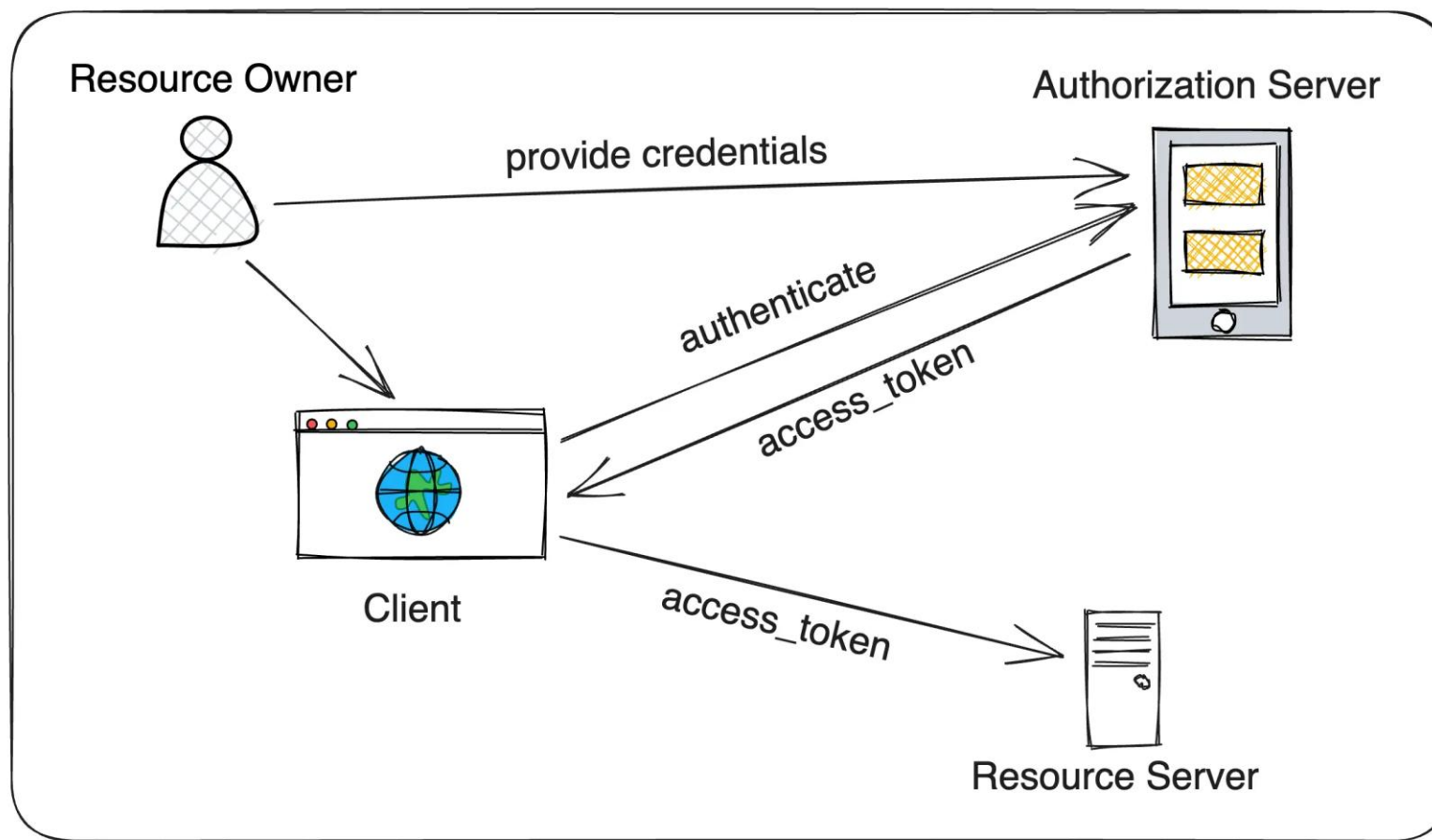
- ☐ 使用TLS（服务器/客户端认证）
- ☐ 基于令牌的认证（OAuth2, JWT）

对于消息代理（Kafka, RabbitMQ）：

- ☐ SSL/TLS用于传输加密
- ☐ 认证用 SASL
- ☐ 用于主题/队列权限的ACL

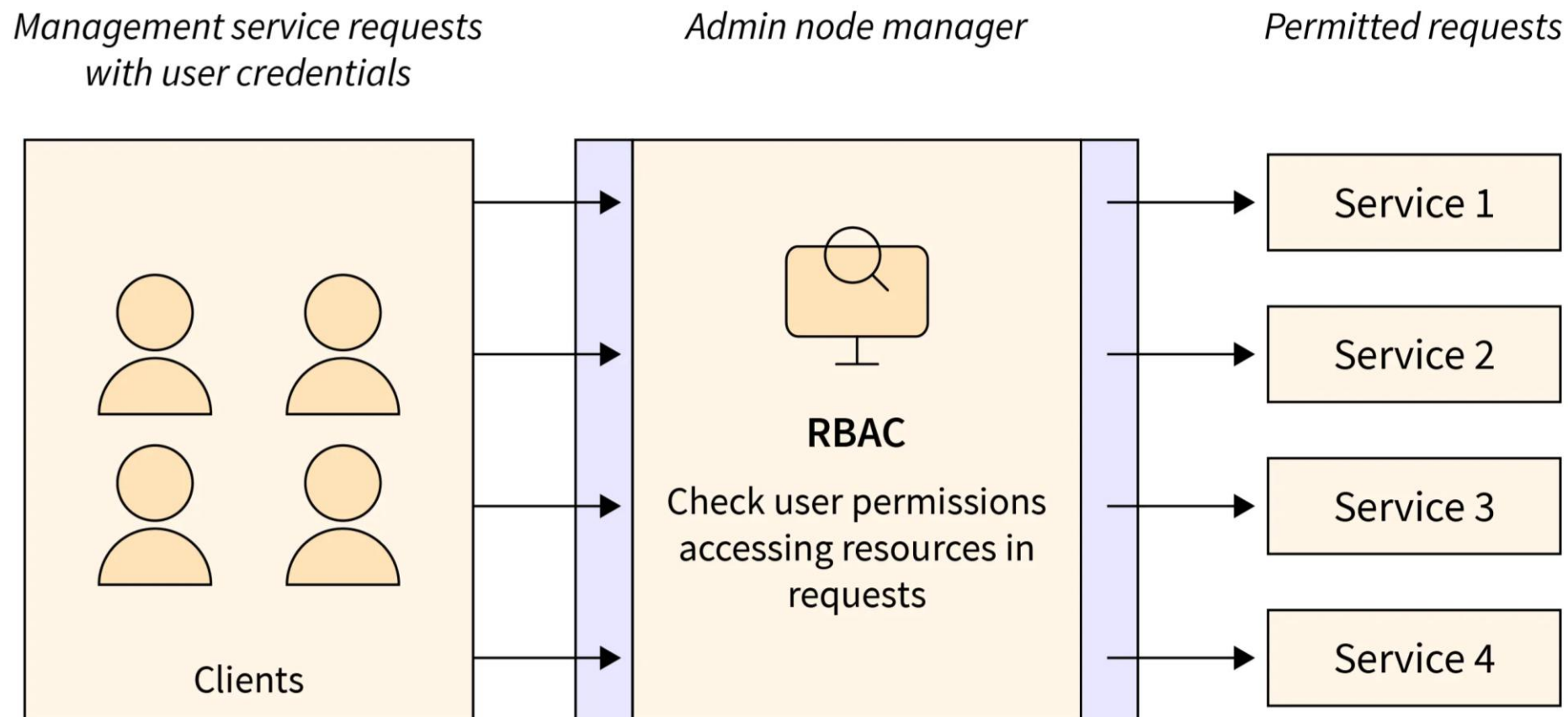
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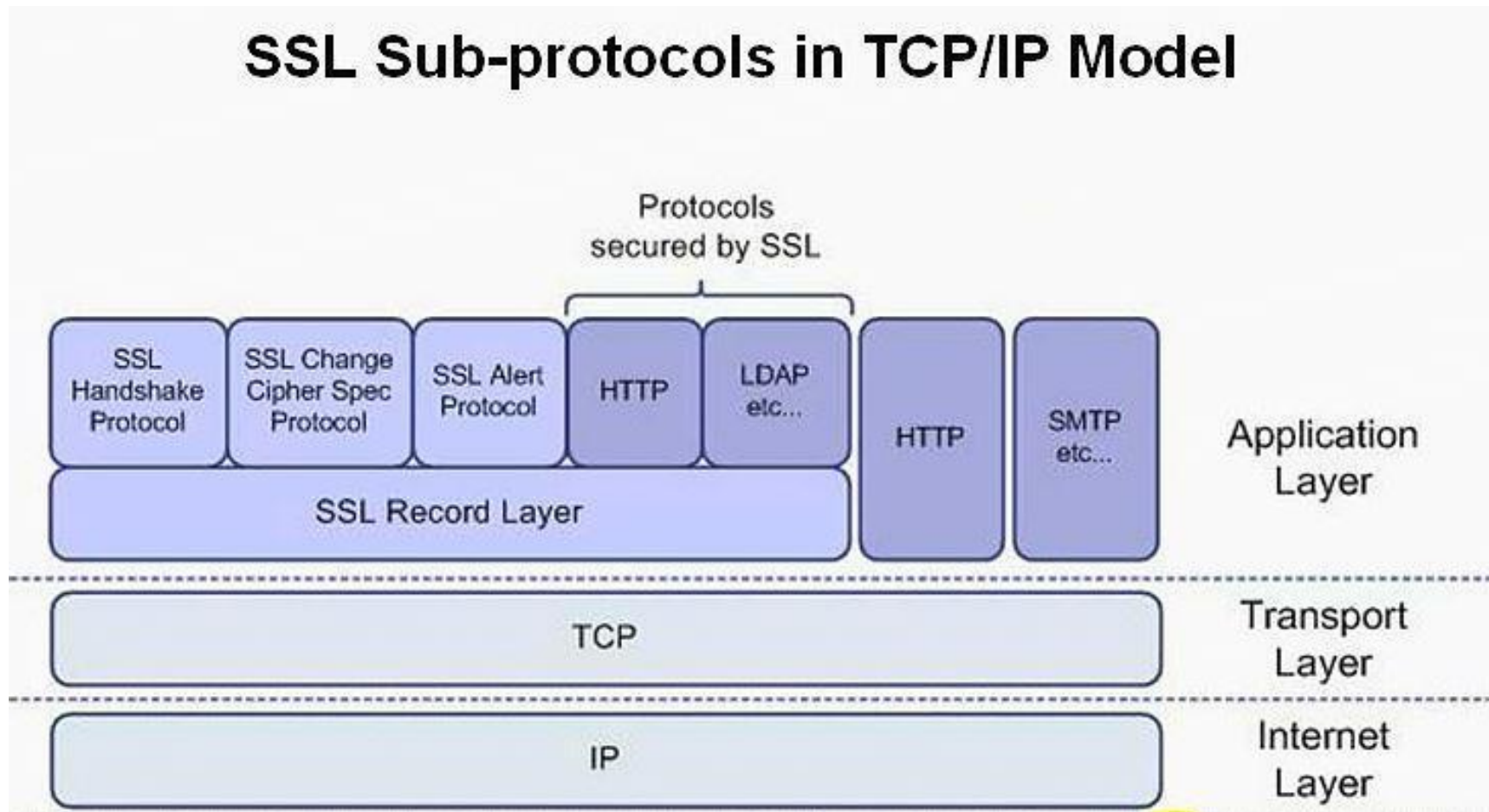
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分布式系统中的通信

Communication in Distributed Systems

选择合适的沟通方式

请提出以下问题：

- ☐ 发件人可以等回复吗？ → RPC
- ☐ 需要缓冲流量突发？ → 消息队列
- ☐ 有没有接收器对活动感兴趣？ → 出版/订阅
- ☐ 高可靠性/一次送达？ → 消息队列

混合架构很常见—例如，命令用 HTTP/RPC，事件用 Kafka

Choosing the Right Communication Mechanism

Ask these questions:

- ☐ Can the sender wait for a reply? → RPC
- ☐ Need to buffer bursts of traffic? → Message queue
- ☐ Many receivers interested in events? → Pub/Sub
- ☐ High reliability / exactly-once delivery? → Message queue

Hybrid architectures are common – e.g., HTTP/RPC for commands, Kafka for events

分布式系统中的通信

Communication in Distributed Systems

Real-World Examples

Microservices: gRPC for synchronous calls + RabbitMQ for async tasks

IoT sensors: MQTT pub/sub (lightweight, unreliable networks)

Data pipelines: Apache Kafka (pub/sub with replayability, ordering)

Financial trading: Low-latency RPC (e.g., Aeron) + persistent queues for audit

现实世界的例子

微服务: gRPC 用于同步调用 + RabbitMQ 用于异步任务

物联网传感器: MQTT 发布/订阅 (轻量级且不可靠的网络)

Data pipelines: Apache Kafka (pub/sub with replayability, ordering)

金融交易: 低延迟RPC (如Aeron) +持续队列用于审计

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Communication in Distributed Systems

Conclusion – Both Are Needed

RPC and message-oriented communication are complementary

RPC: simple, low-latency request-reply

Message passing: resilient, decoupled, scalable

Security must be applied end-to-end

Key takeaway: Choose the right tool for the communication pattern

结论—两者都是必需的

RPC和面向消息的通信是互补的

RPC: 简单、低延迟的请求-回复

消息传递: 有弹性、解耦、可扩展

安全措施必须端到端应用

关键点: 选择适合沟通模式的工具

分布式系统中的通信

Communication in Distributed Systems

Further Resources

- ❑ gRPC official docs – grpc.io
- ❑ Kafka: The Definitive Guide (O'Reilly)
- ❑ “Designing Data-Intensive Applications” – Martin Kleppmann (Chapters 4, 11)
- ❑ MQTT specification – mqtt.org
- ❑ Open source examples: GitHub – [grpc-examples](#), [kafka-examples](#)

更多资源

- ❑ gRPC官方文档 – grpc.io
- ❑ 《卡夫卡：权威指南》（奥莱利）
- ❑ 《设计数据密集型应用》—Martin Kleppmann（第4、11章）
- ❑ MQTT 规范 – mqtt.org
- ❑ 开源示例：GitHub – [grpc-examples](#), [kafka-examples](#)

分布式系统中的命名

Naming in Distributed Systems

What Is Naming?

Name: An identifier for a resource (e.g., file, node, service)

Address: Location of a resource (e.g., IP address)

Identifier: A name that uniquely and permanently refers to a resource

Pure name: No location info (e.g., UUID) – also called opaque

Impure name: Contains location/context (e.g., alice@example.com)

什么是命名？

名称：资源的标识符（例如文件、节点、服务）

地址：资源的位置（例如，IP地址）

标识符：唯一且永久地指向资源的名称

纯名称：无位置信息（例如 UUID）——也称为不透明

不纯名称：包含位置/上下文（例如，alice@example.com）

分布式系统中的命名

Naming in Distributed Systems

Key Concepts & Goals

Name space: Set of all valid names

Naming domain: Names managed by a single authority

Goals of naming in distributed systems:

- ☐ Uniqueness (no collisions)
- ☐ Location-independence (resource can move)
- ☐ Scalability (billions of names)
- ☐ Fault tolerance (no single point of failure)

关键概念与目标

命名空间: 所有有效名称的集合

命名域名: 由单一权威管理的名称

分布式系统中命名的目标:

- ☐ 唯一性 (无碰撞)
- ☐ 位置无关性 (资源可以移动)
- ☐ 可扩展性 (数十亿个名字)
- ☐ 容错 (无单点故障)

分布式系统中的命名

Naming in Distributed Systems

Flat Naming – Overview

Definition: Names are random-looking, no inherent hierarchy

Examples: UUID (Universally Unique Identifier), object IDs in BitTorrent (SHA-1 hash)

Properties:

- ☐ Simple, no delegation needed
- ☐ Supports strong decentralization
- ☐ Requires a mapping service (e.g., DHT) to resolve name → address

公寓命名——概述

定义： 名字看起来很随机，没有固有的等级

示例： UUID（通用唯一标识符）、BitTorrent中的对象ID（SHA-1哈希）

房产：

很简单，不需要委托

支持强力的去中心化

需要映射服务（例如 DHT）来解析名称→地址

分布式系统中的命名

Naming in Distributed Systems

Flat Naming – Security Challenges & Solutions

Challenges:

- ☐ Spoofing (fake name-to-address mapping)
- ☐ Tampering (modifying stored mappings)
- ☐ Unauthorized access

Solutions:

- ☐ Cryptographic hash functions (e.g., SHA-256) for unique IDs
- ☐ Public Key Infrastructure (PKI) for authentication
- ☐ Blockchain (e.g., Ethereum Name Service) – tamper-proof, decentralized
- ☐ Distributed Hash Tables (DHTs) with secure routing
- ☐ Digital signatures & encryption for integrity/confidentiality

公寓命名—安全挑战与解决方案

挑战:

- ☐ 伪造（假姓名到地址映射）
- ☐ 篡改（修改存储映射）
- ☐ 未经授权的访问

解决方案:

- ☐ 唯一ID的密码学哈希函数（例如SHA-256）
- ☐ 用于认证的公钥基础设施（PKI）
- ☐ 区块链（例如以太坊名称服务）—防篡改，去中心化
- ☐ 具有安全路由的分布式哈希表（DHT）
- ☐ 数字签名与加密以实现完整性/机密性

分布式系统中的命名

Naming in Distributed Systems

Chord

Consider the organization of many nodes into a logical ring

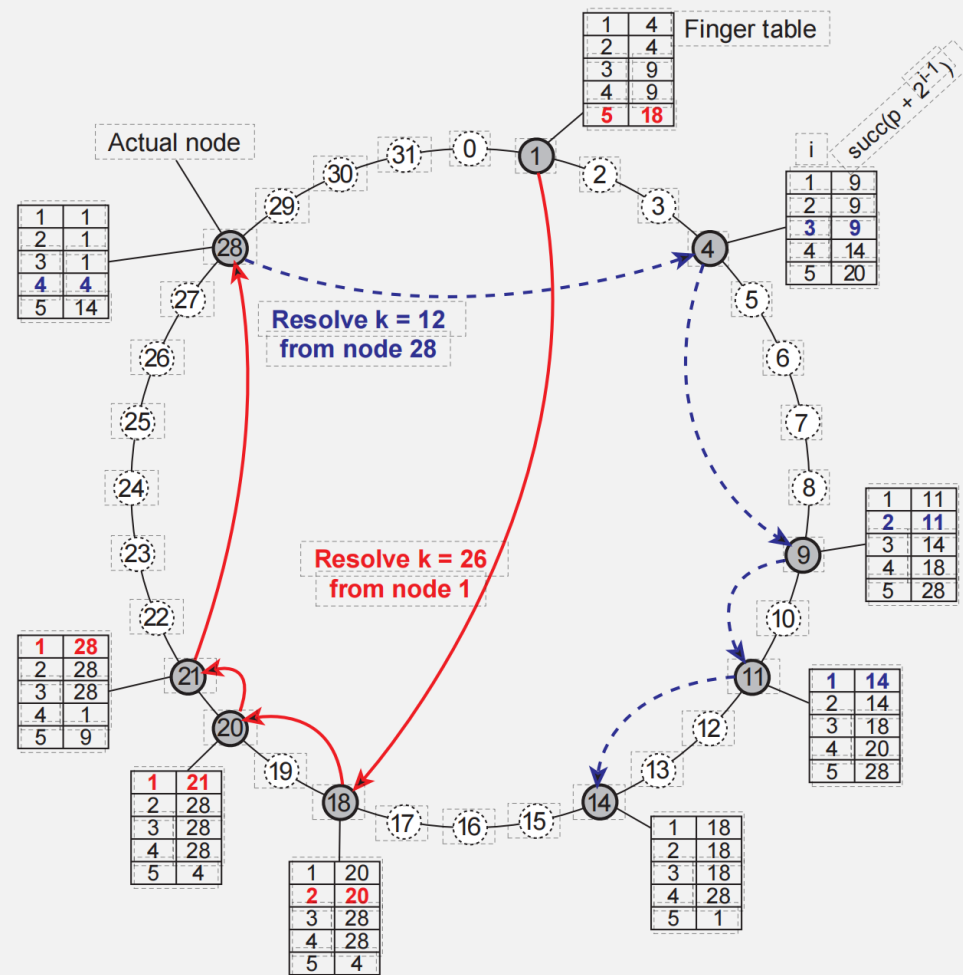
Each node is assigned a random m-bit identifier.

Every entity is assigned a unique m-bit key.

Entity with key k falls under jurisdiction of node with smallest

id $\geq k$ (called its successor).

Resolve k=21 from 28



分布式系统中的命名

Naming in Distributed Systems

Hierarchical Naming – Structure

Tree hierarchy:

- ❑ Root (unnamed starting point)
- ❑ Nodes represent domains / subdomains
- ❑ Leaves represent individual resources

Delegation: Each level delegates authority to lower levels (e.g., ICANN → registry → DNS operator)

层级命名—结构

树状结构:

- ❑ 根（未命名起始点）
- ❑ 节点表示域/子域
- ❑ 叶子代表个体资源

委托: 每个层级将权限委托给较低层级（例如，ICANN注册局→DNS运营商→

分布式系统中的命名

Naming in Distributed Systems

Domain Name System (DNS) – The Classic

Example

Hierarchical namespace: com → example → www

Components:

- ❑ Resolvers (client side)
- ❑ Name servers: root, Top-Level Domain (TLD), authoritative

Resolution process:

Resolver asks root → root directs to TLD server →
TLD directs to authoritative server → returns IP

域名系统（DNS）——经典示例

层级命名空间：com → www → 示例

组成部分：

- ❑ 解析器（客户端）
- ❑ 名称服务器：根服务器，顶级域名（TLD），权威服务器

解决流程：

解析器请求root，root→指向顶级域名服务器→TLD
指向权威服务器→返回IP

分布式系统中的命名

Naming in Distributed Systems

DNS in Action – Code Example DNS实务—代码示例

```
import socket

domain = "cs.bjut.edu.cn"
ip_address = socket.gethostbyname(domain)
print(f"{domain} -> {ip_address}")
```

This uses the system's configured DNS resolver transparently

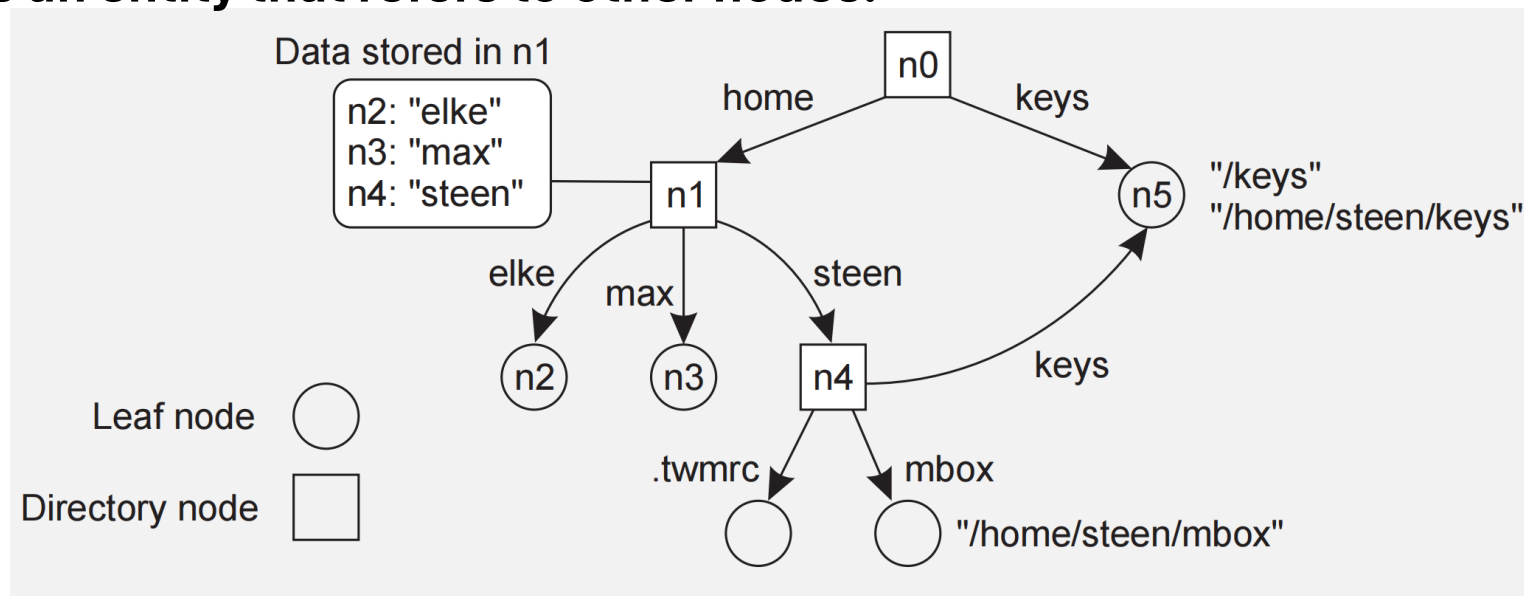
Behind the scenes: recursive queries, caching, TTL

分布式系统中的命名

Naming in Distributed Systems

A graph in which a leaf node represents a (named) entity.

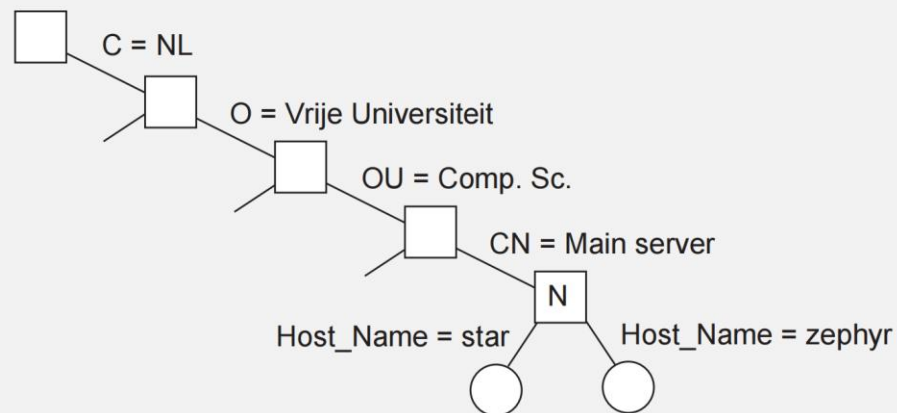
A directory node is an entity that refers to other nodes.



分布式系统中的命名

Naming in Distributed Systems

Attribute-based naming



Attribute	Value
Country	NL
Locality	Amsterdam
Organization	Vrije Universiteit
OrganizationalUnit	Comp. Sc.
CommonName	Main server
Host_Name	star
Host_Address	192.31.231.42

Attribute	Value
Country	NL
Locality	Amsterdam
Organization	Vrije Universiteit
OrganizationalUnit	Comp. Sc.
CommonName	Main server
Host_Name	zephyr
Host_Address	137.37.20.10

分布式系统中的命名

Naming in Distributed Systems

Hierarchical Naming – Pros and Cons

Advantages:

- ☐ Human-readable names
- ☐ Scalable administration (delegation)
- ☐ Efficient resolution with caching

Disadvantages:

- ☐ Root servers are potential bottlenecks
- ☐ Centralised trust (root is controlled by ICANN)
- ☐ Hard to support highly dynamic resource mobility

层级命名—利弊

优势:

- ☐ 人类可读的名字
- ☐ 可扩展管理（委托）
- ☐ 通过缓存实现高效解析

缺点:

- ☐ 根服务器可能是瓶颈
- ☐ 中心式信托（根由ICANN控制）
- ☐ 很难支持高度动态的资源流动

分布式系统中的命名

Naming in Distributed Systems

Comparison – Flat vs. Hierarchical Naming 命名

比较——平面命名与层级

Feature	Flat Naming	Hierarchical Naming
Structure	No hierarchy	Tree
Human-friendliness	Low (e.g., UUID)	High (e.g., DNS names)
Decentralisation	High (e.g., DHT, blockchain)	Moderate (root is central)
Resolution cost	$O(\log N)$ in DHT	$O(\text{depth})$ with caching
Typical use	P2P networks, object stores	Web, email, file systems

分布式系统中的命名

Naming in Distributed Systems

Best Practices for Naming in Distributed Systems

- ❑ Use late binding – resolve names at runtime, avoid hardcoding addresses
- ❑ Redundancy & replication – multiple name servers for fault tolerance
- ❑ Apply cryptography – especially for flat naming (signatures, hashes)
- ❑ Cache with TTL – balance performance and freshness
- ❑ Plan for mobility – use indirection (e.g., forwarding pointers, home agents)

分布式系统命名的最佳实践

- ❑ 使用晚绑定—在运行时解析名称，避免硬编码地址
- ❑ 冗余与复制—多名称服务器以实现容错
- ❑ 应用密码学—尤其是针对平面命名（签名、哈希）
- ❑ TTL缓存—平衡性能和新鲜度
- ❑ 为流动性做规划—使用间接性（例如，转发指针、家庭代理）

分布式系统中的命名

Naming in Distributed Systems

Real-World Applications

- ❑ IoT device IDs – flat naming via EUI-64 or cryptographic IDs
- ❑ Cloud resource management – hierarchical names (e.g., AWS ARN)
- ❑ P2P networks – DHTs (Kademlia, Chord) for flat key-based routing
- ❑ Content Delivery Networks (CDNs) – DNS-based request routing
- ❑ Distributed file systems – HDFS, Ceph (hybrid naming)

实际应用

- ❑ 物联网设备ID—通过EUI-64或加密ID进行平面命名
- ❑ 云资源管理—层级名称（例如，AWS ARN）
- ❑ P2P网络—DHT（Kademlia、Chord）用于基于平坦密钥的路由
- ❑ 内容分发网络（CDN）—基于DNS的请求路由
- ❑ 分布式文件系统—HDFS，Ceph（混合命名）

分布式系统中的命名

Naming in Distributed Systems

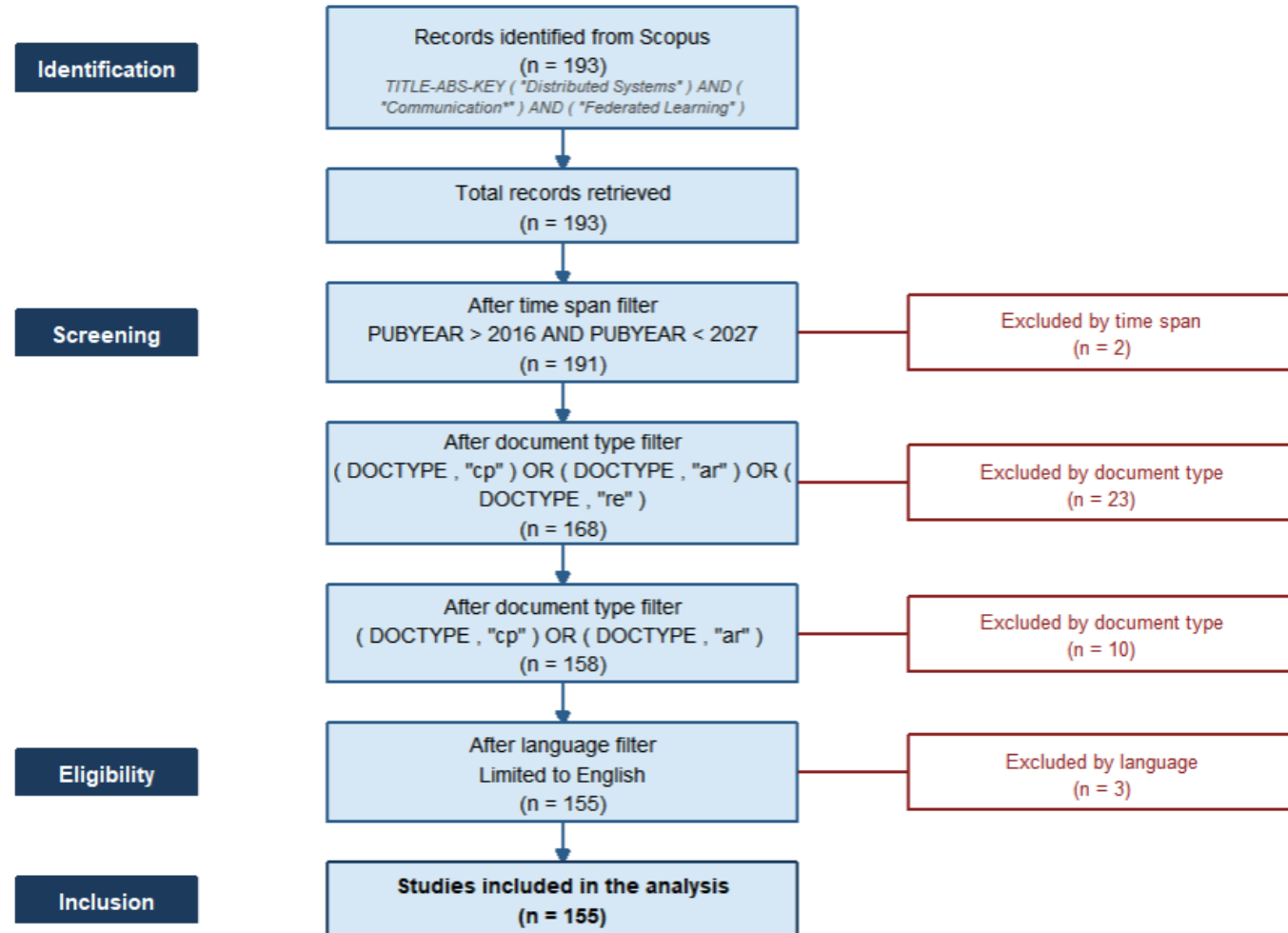
摘要

- ❑ 命名对于定位和访问分布式资源至关重要
- ❑ 平面命名 = 简单、去中心化，需要安全映射服务
- ❑ 层级命名 = 结构化、人类可读的委托
- ❑ DNS 是最广泛的层级命名系统
- ❑ 安全性（加密、签名、区块链）在这两种方法中都至关重要
- ❑ 选择基于：人类可读性、可扩展性、移动性、信任模型

Summary

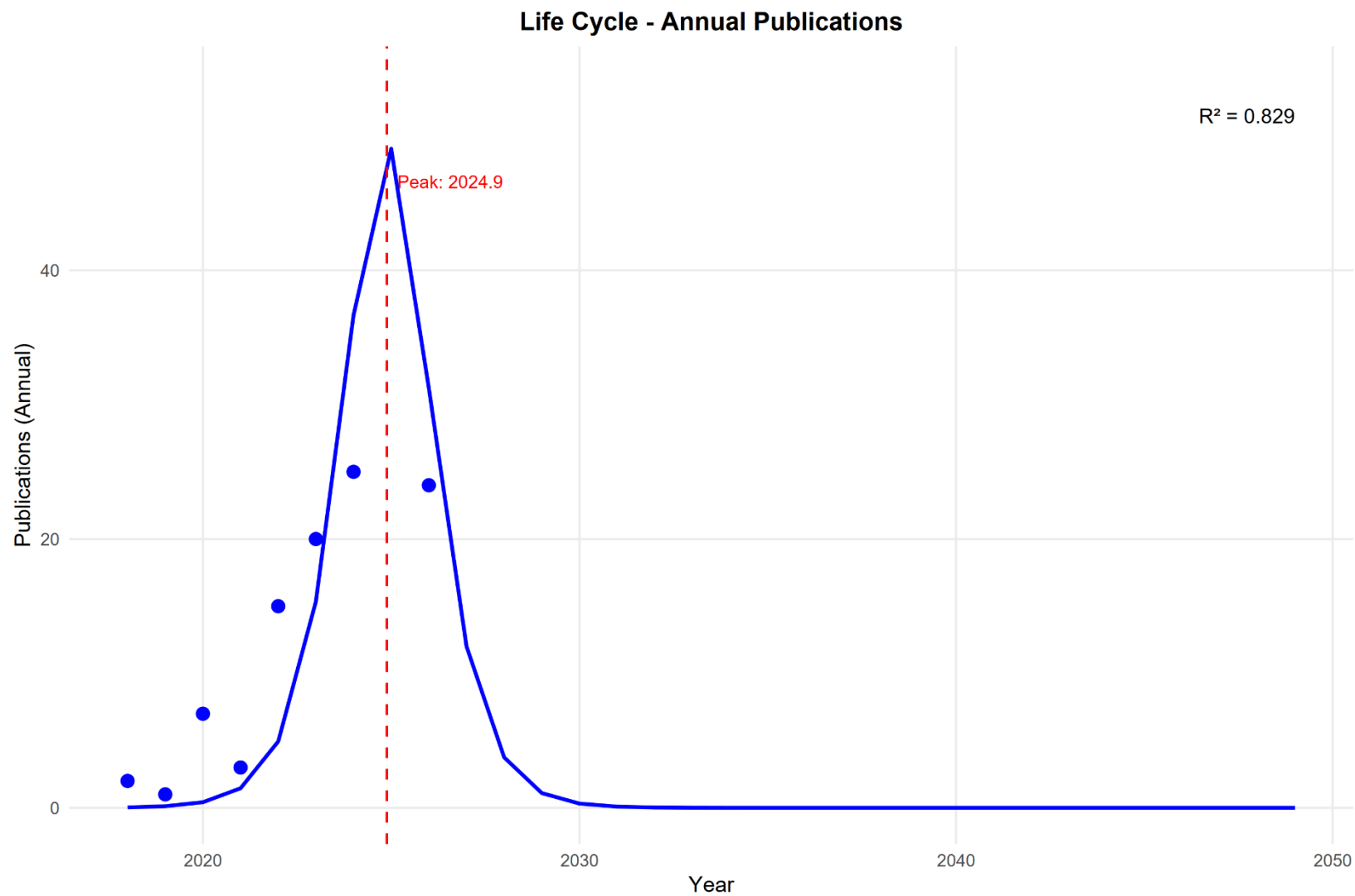
- ❑ Naming is essential for locating and accessing distributed resources
- ❑ Flat naming = simple, decentralized, needs secure mapping service
- ❑ Hierarchical naming = structured, human-readable, delegation
- ❑ DNS is the most widespread hierarchical naming system
- ❑ Security (encryption, signatures, blockchain) is critical in both approaches
- ❑ Choose based on: human readability, scalability, mobility, trust model

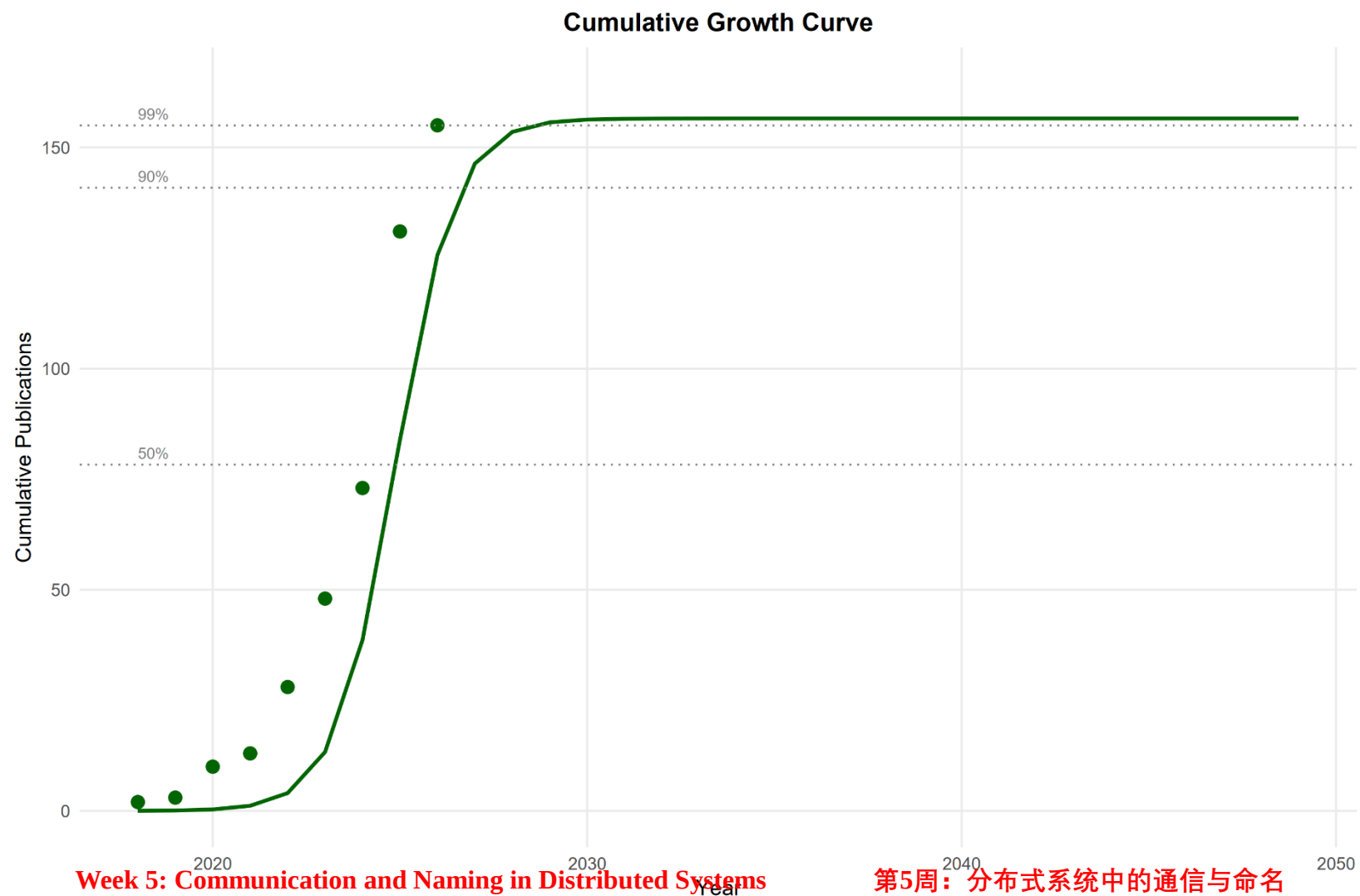
PRISMA Flow Diagram (adapted for bibliometric studies)



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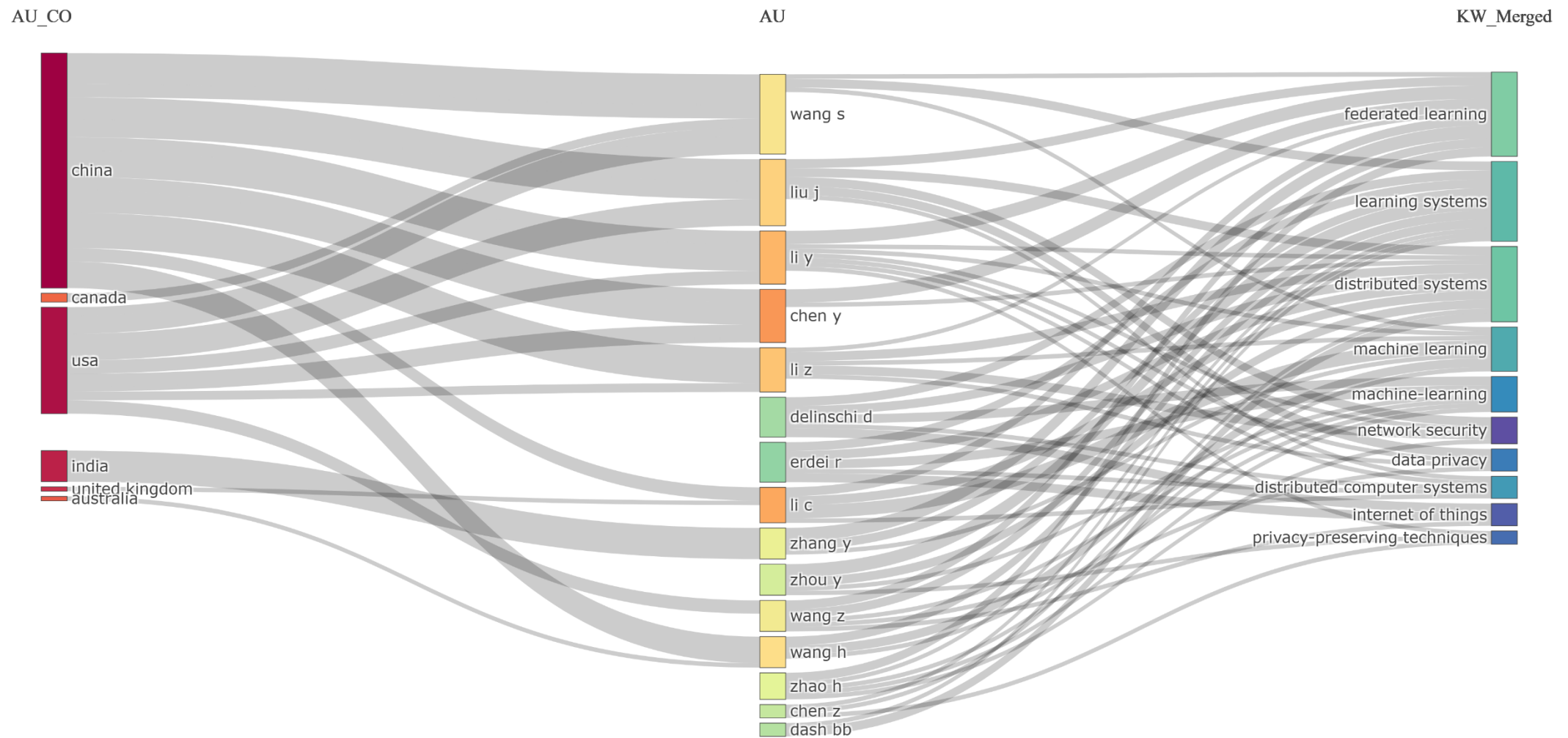


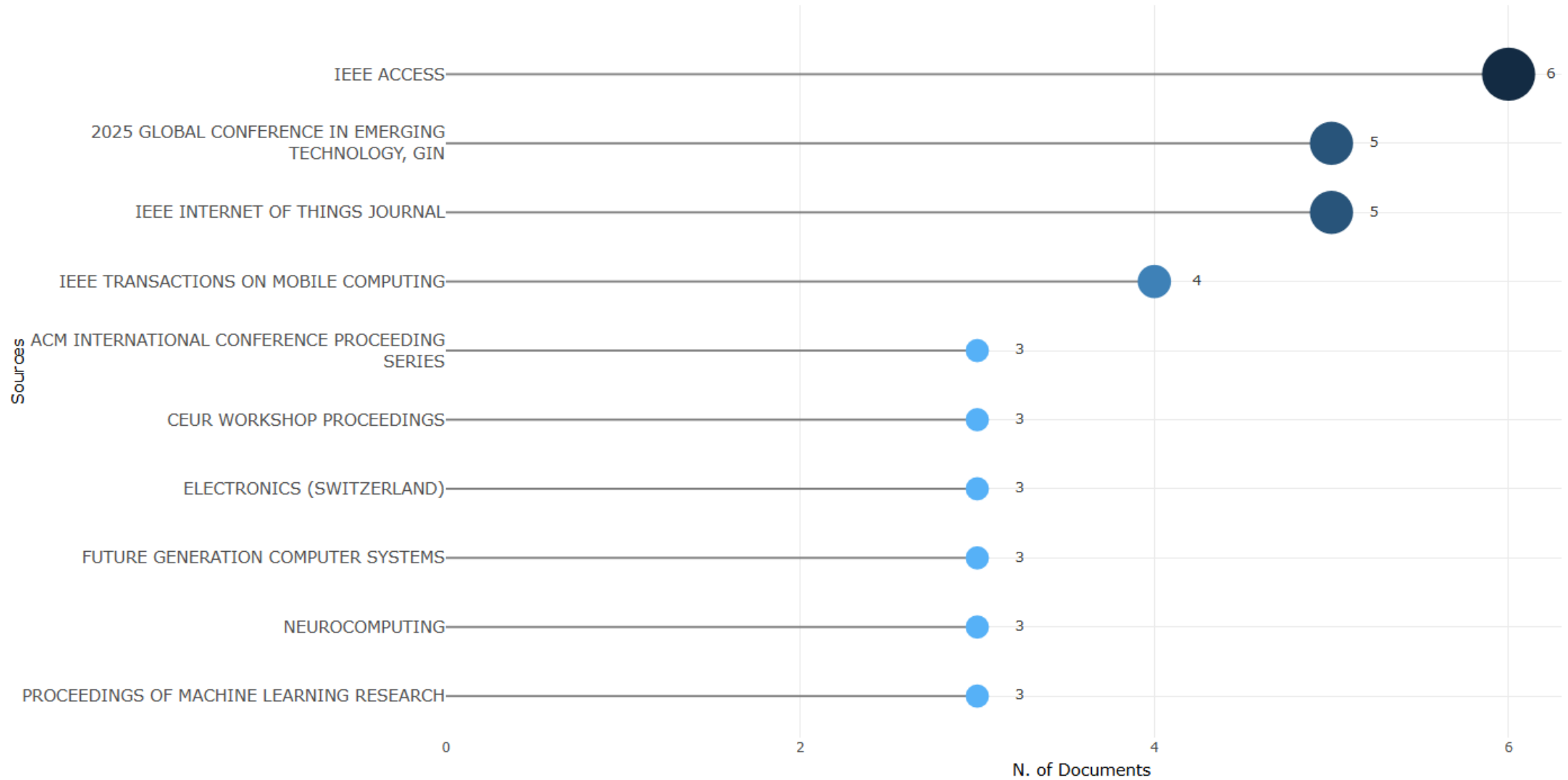




Week 5: Communication and Naming in Distributed Systems

第5周：分布式系统中的通信与命名



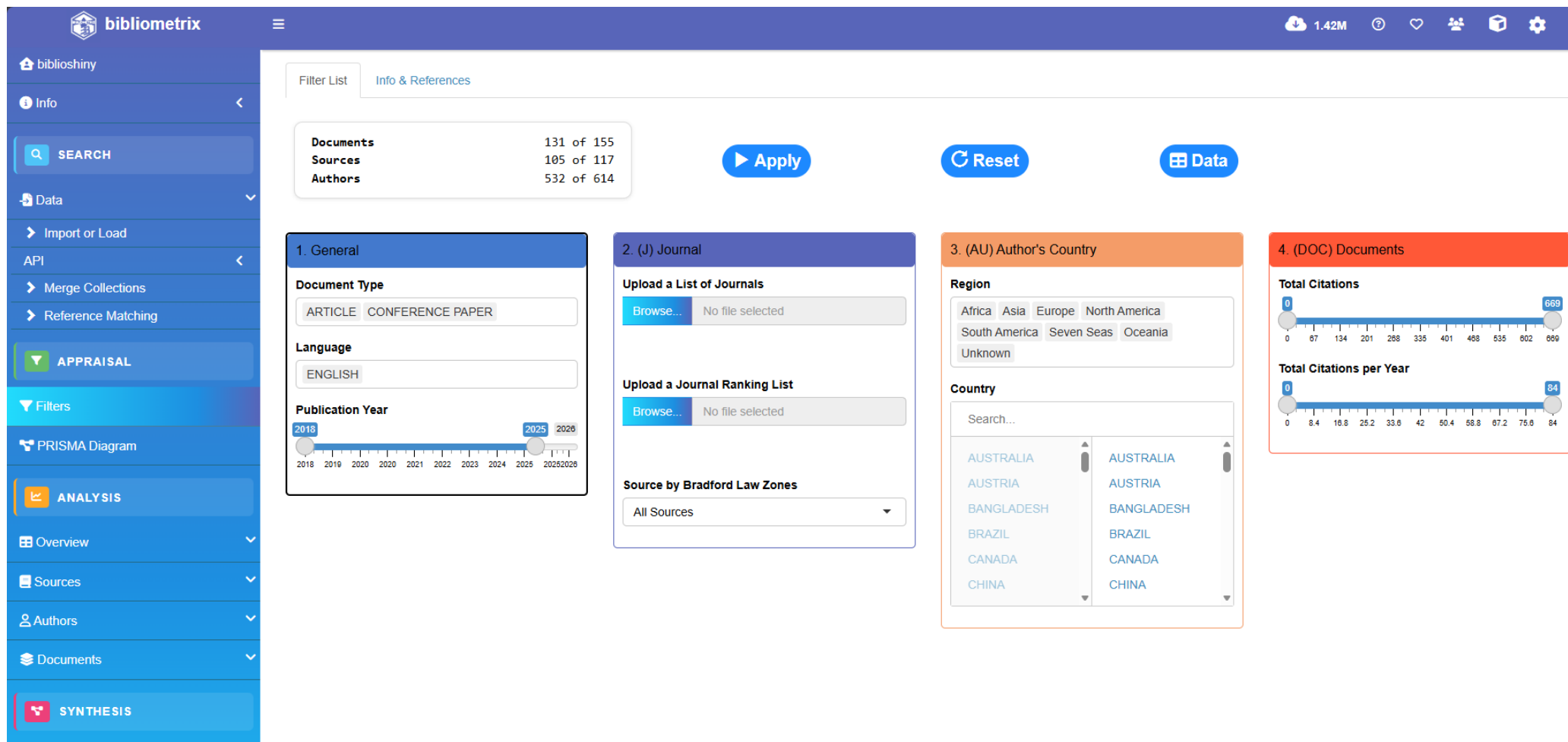


Country	Freq
Filter...	Filter...
CHINA	104
USA	68
INDIA	64
UK	13
ITALY	12
GERMANY	11
AUSTRALIA	10
CANADA	10
SOUTH KOREA	9
SAUDI ARABIA	7

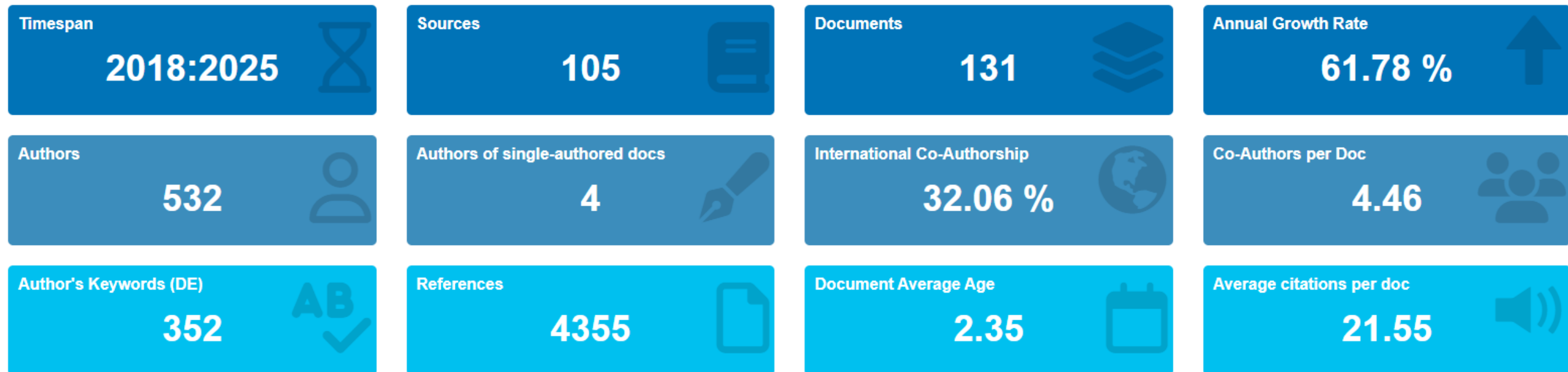
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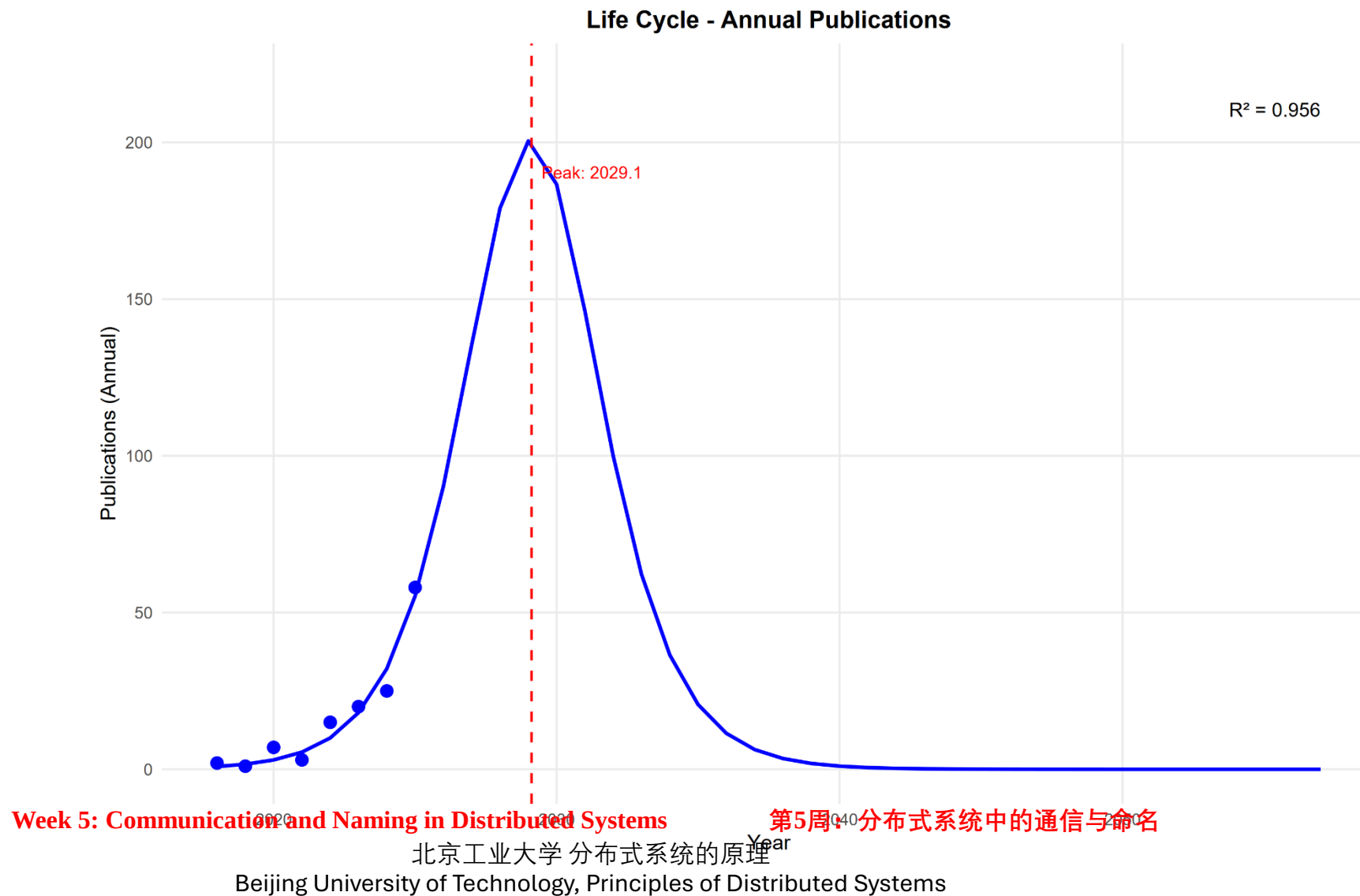
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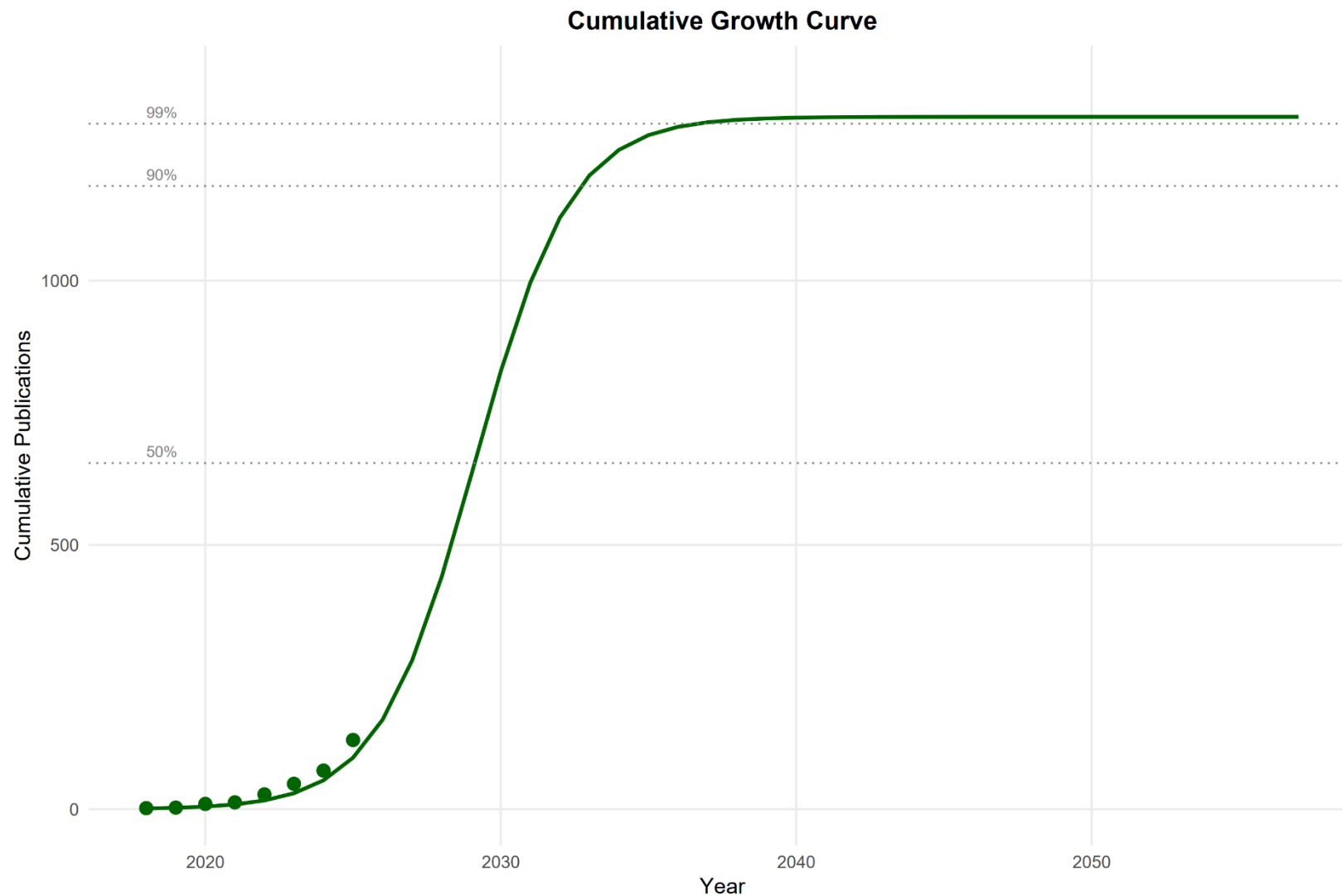
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Final takeaway for Week 2 第二周总结



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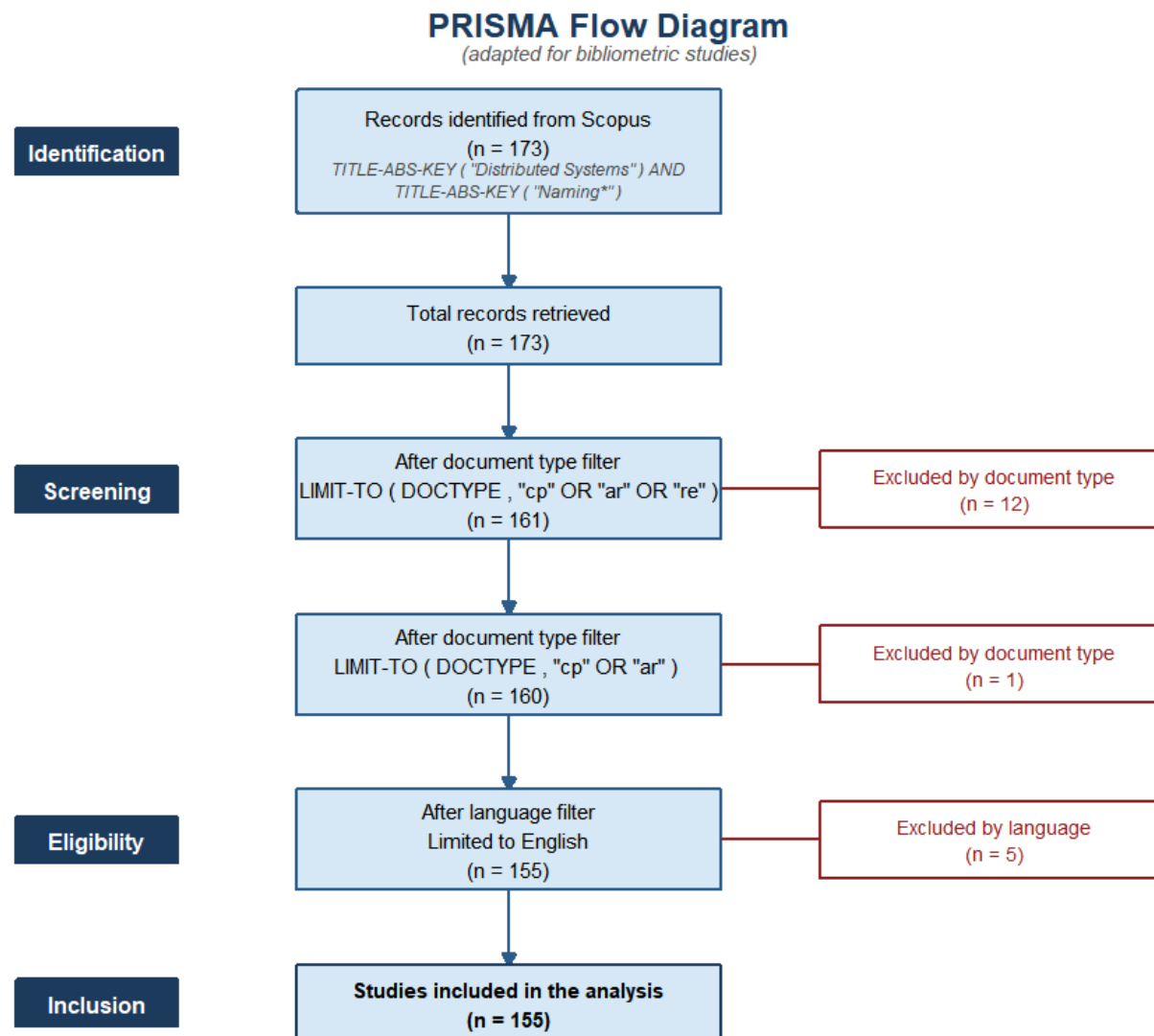
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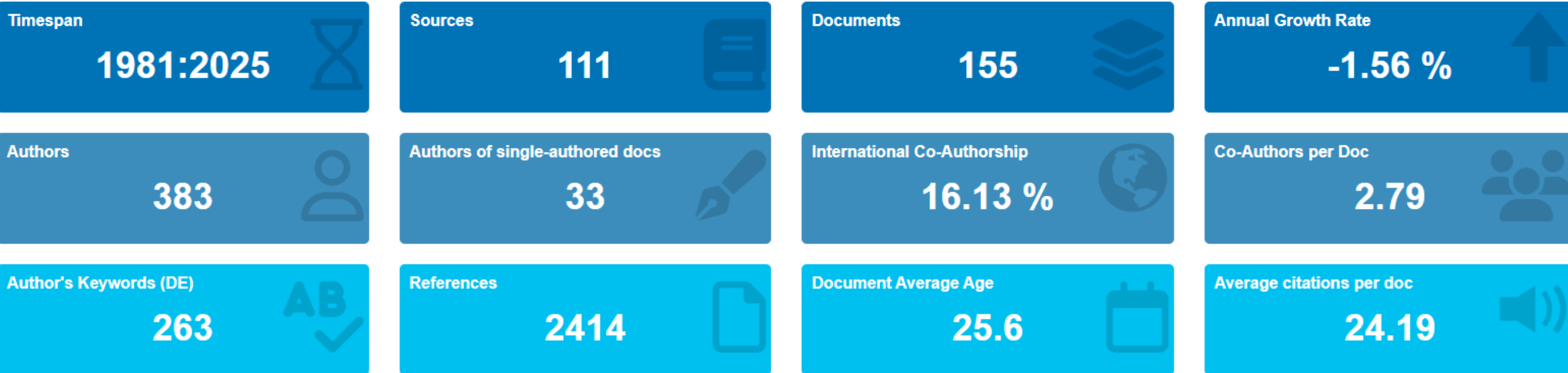
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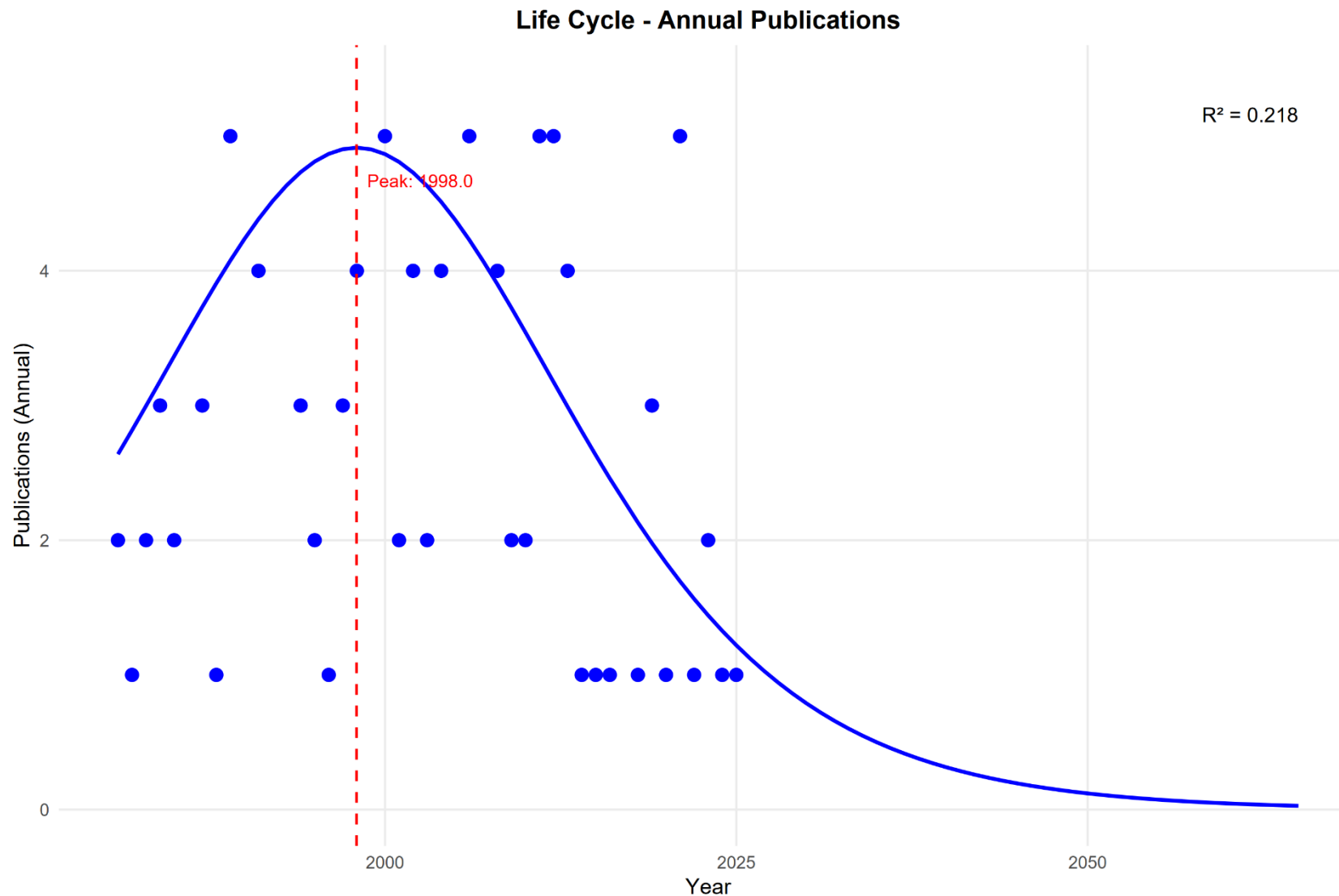
Naming in Distributed Systems 第二周总结



Naming in Distributed Systems 第二周总结



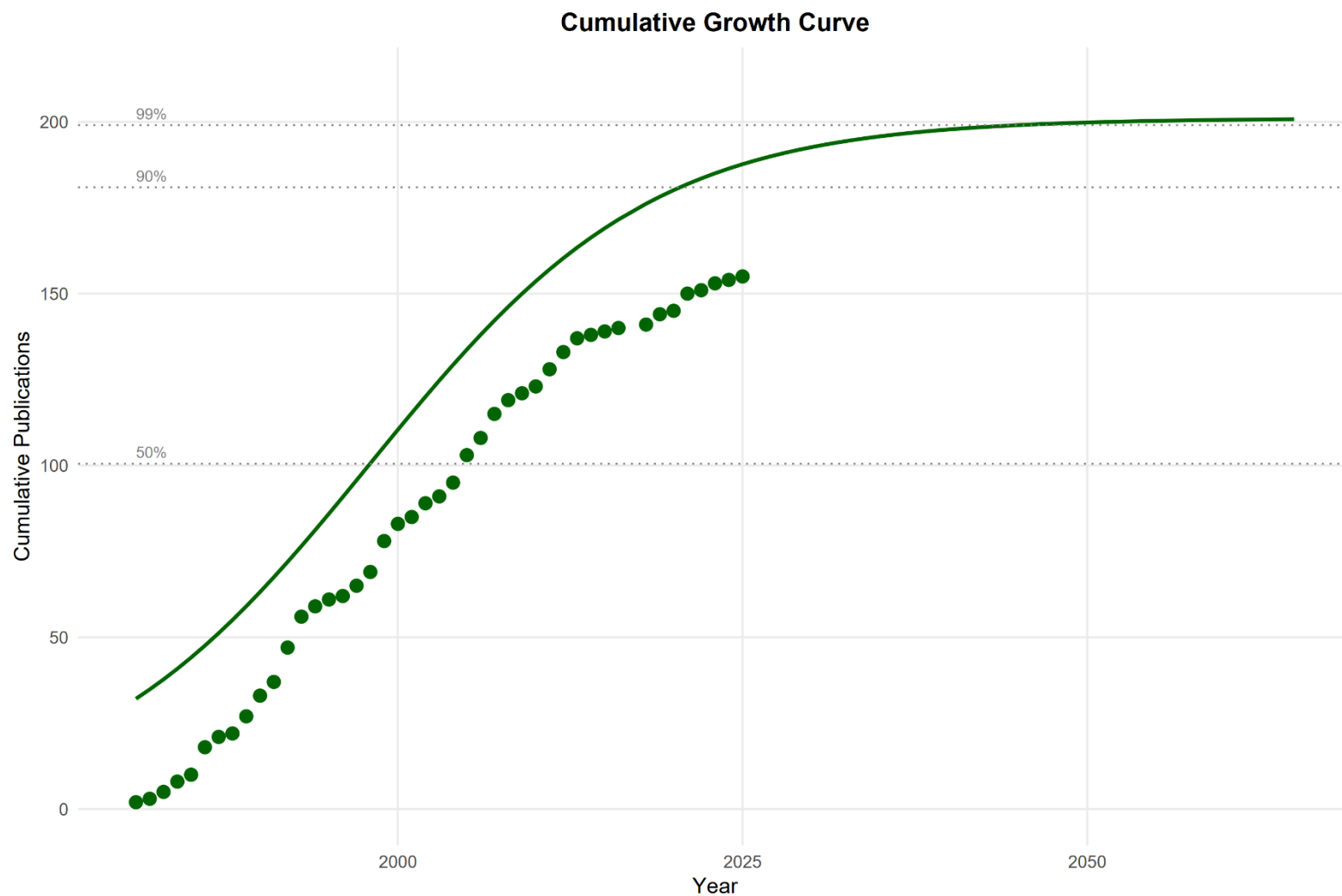
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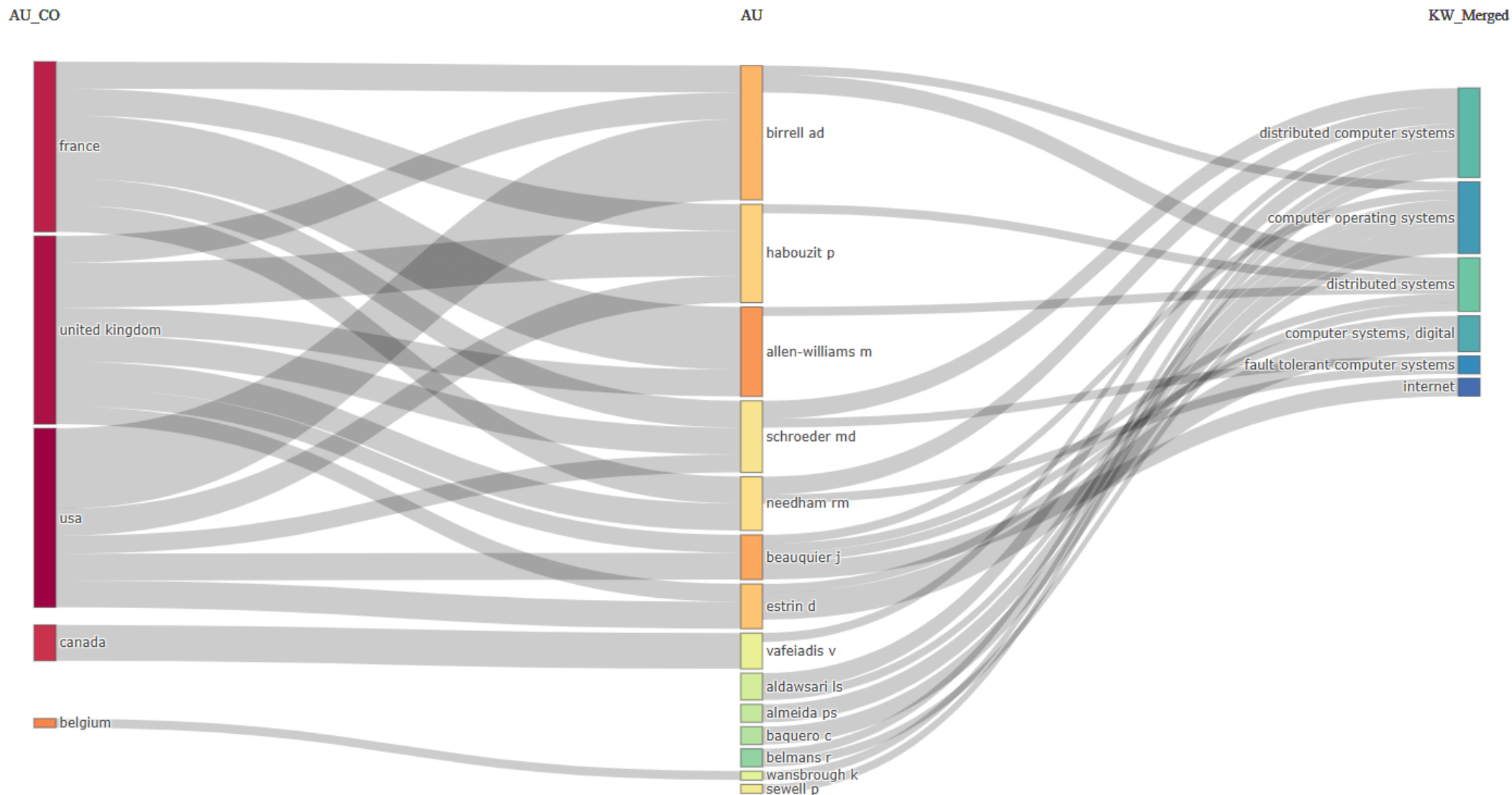
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Final takeaway for Week 2 第二周总结



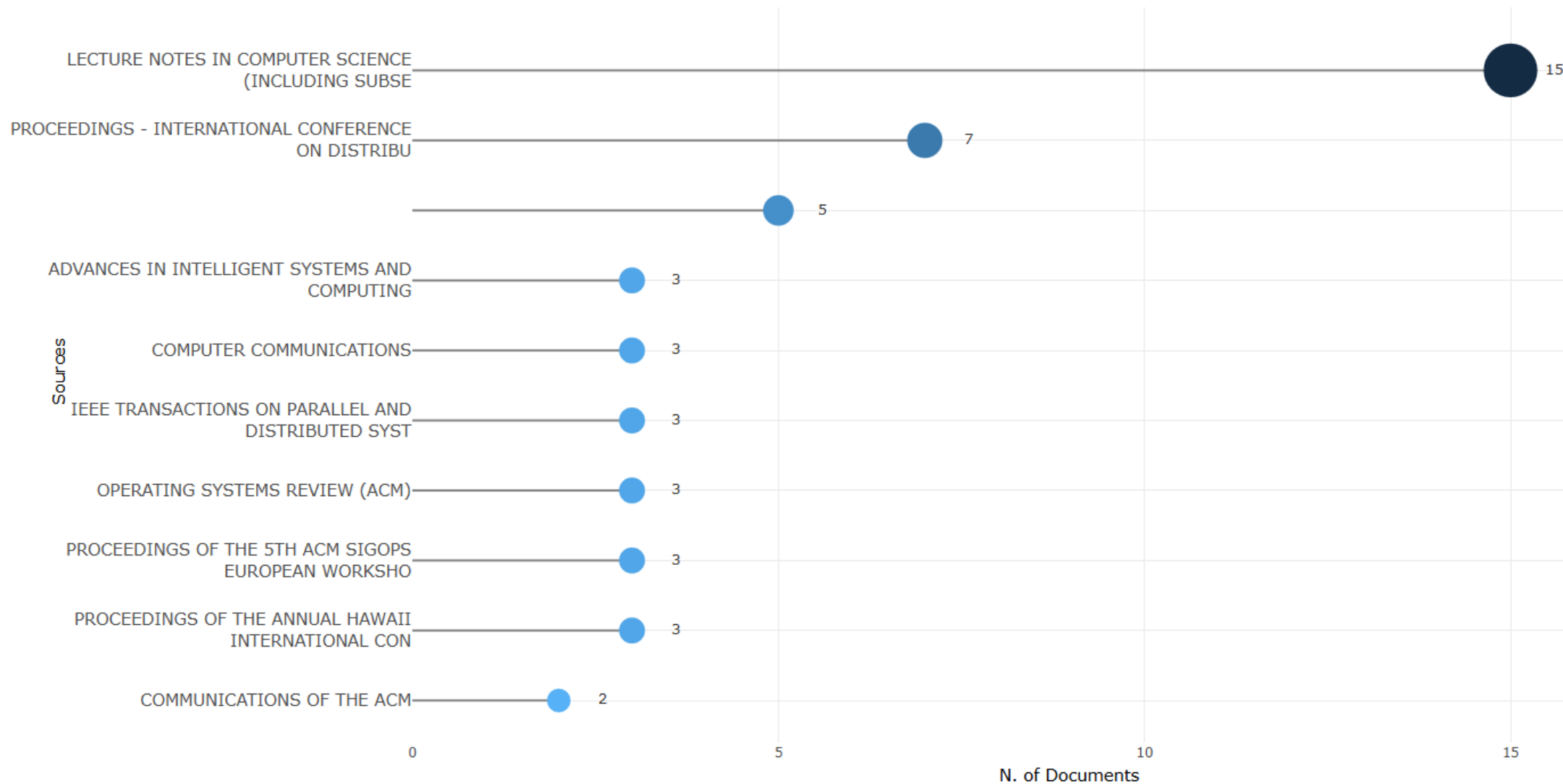
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Country	Freq
Filter...	Filter...
USA	101
UK	27
FRANCE	19
CANADA	14
ITALY	14
AUSTRALIA	8
GERMANY	8
CHINA	7
JAPAN	4
BELGIUM	3

Final takeaway for Week 2 第二周总结



Week 5: Communication and Naming in Distributed Systems

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Clarivate

English Products

Web of Science™ Smart Search Advanced Search Research Assistant

Sign In Register

Search Results for "Distributed System*" (Topic) AND "Communication*" (Topic) A...

91 results from Web of Science Core Collection for:

"Distributed System*" (Topic) and "Communication*" (Topic) and "Federated Learning" (Topic)

+ Add Keywords Quick add keywords: hierarchical federated learning federated learning distributed systems federated learning fl di

91 Documents You may also like... Analyze Results Citation Report Create Alert

Refine results Export Refine

Search within results...

Quick Filters

- ☐ Review Article 4
- ☐ Early Access 3
- ☐ Open Access 33
- ☐ Enriched Cited References 31

Publication Years

Show Final Publication Year

2026 16

0/91 Add To Marked List Export

Sort by Relevance 1 of 2

1 On the Optimal Error Exponent of Type-Based Distributed Hypothesis Testing 32 References

Tong, XY; Xu, XX and Huang, SL
Oct 2023 | ENTROPY 25(10)

Enriched Cited References

Distributed hypothesis testing (DHT) has emerged as a significant research area, but the information-theoretic optimality of coding strategies is often typically hard to address. This paper studies the DHT problems under the type-based setting, which is requested from the popular federated learning methods. Specifically, two communication models ε ... Show more

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Related records

2 Federated learning in cloud-edge-fog architectures: Enhancing privacy, efficiency, and scalability 156 References

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Scopus

Search

Sources

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FS

Advanced query

Search within
Article title, Abstract, Keywords

Search documents *
"Distributed System*"

AND

Search within
Article title, Abstract, Keywords

Search documents
"Communication*"

AND

Search within
Article title, Abstract, Keywords

Search documents
"Federated Learning"

Save search

Set search alert

Add search field

Reset

Search

Documents Preprints Secondary documents

195 documents found

Analyze results

Refine search

Search within results

Filters

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	Document title	Authors	Source	Year	Citations
1	Federated edge intelligence for carbon emission forecasting using deep neural networks: a privacy-preserving approach	Li, Y., Fard, K.G.	Journal of Cloud Computing , 15(1), 57	2026	0
	Show abstract	Find it!	View at Publisher	Related documents	
2	Research on privacy protection and performance optimization of 6G communication network based on the fusion of federated	Zhao, H.	Discover Artificial Intelligence	2026	0

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The screenshot displays the Bibliometrix web application interface. The top navigation bar includes the 'bibliometrix' logo, a hamburger menu, and utility icons for download (1.42M), help, favorites, users, settings, and power. The left sidebar contains navigation links: 'biblioshiny', 'Info', 'SEARCH', 'Data', 'APPRAISAL', 'Filters', 'PRISMA Diagram', 'ANALYSIS', 'Overview', 'Main Information', 'Annual Scientific Production', 'Average Citations per Year', 'Life Cycle', 'Three-Field Plot', 'Sources', and 'Authors'.

The main content area is divided into several sections:

- Filter List**: Includes 'Info & References' and a summary of selected items: Documents (170 of 193), Sources (136 of 149), and Authors (680 of 743). Buttons for 'Apply', 'Reset', and 'Data' are present.
- 1. General**: Contains filters for Document Type (ARTICLE, ARTICLE; EARLY ACCESS, CONFERENCE PAPER, PROCEEDINGS PAPER), Language (ENGLISH, CHINESE), and Publication Year (2013 to 2026).
- 2. (J) Journal**: Includes sections for 'Upload a List of Journals' and 'Upload a Journal Ranking List', both with 'Browse...' buttons and 'No file selected' status. A 'Source by Bradford Law Zones' dropdown is set to 'All Sources'.
- 3. (AU) Author's Country**: Features a 'Region' filter (Africa, Asia, Europe, North America, South America, Seven Seas, Oceania, Unknown) and a 'Country' list (AUSTRALIA, AUSTRIA, BANGLADESH, BRAZIL, CANADA, CHINA).
- 4. (DOC) Documents**: Displays two horizontal sliders for 'Total Citations' (0 to 489) and 'Total Citations per Year' (0 to 62).

Week 5: Communication and Naming in Distributed Systems

第5周：分布式系统中的通信与命名

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The main content area is divided into several sections:

- Filter List**: Includes 'Info & References' and a summary table:

Category	Count
Documents	165 of 193
Sources	132 of 149
Authors	665 of 743

Buttons for 'Apply', 'Reset', and 'Data' are present.
- 1. General**:
 - Document Type**: ARTICLE, ARTICLE; EARLY ACCESS, CONFERENCE PAPER, PROCEEDINGS PAPER.
 - Language**: ENGLISH.
 - Publication Year**: A range slider from 2013 to 2026, with markers at 2013, 2017, and 2026.
- 2. (J) Journal**:
 - Upload a List of Journals**: Browse... No file selected.
 - Upload a Journal Ranking List**: Browse... No file selected.
 - Source by Bradford Law Zones**: All Sources.
- 3. (AU) Author's Country**:
 - Region**: Africa, Asia, Europe, North America, South America, Seven Seas, Oceania, Unknown.
 - Country**: Search... with a list of countries including AUSTRALIA, AUSTRIA, BANGLADESH, BRAZIL, CANADA, and CHINA.
- 4. (DOC) Documents**:
 - Total Citations**: A range slider from 0 to 489.
 - Total Citations per Year**: A range slider from 0 to 62.

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第5周：分布式系统中的通信与命名

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The screenshot displays the Bibliometrix web application interface. The top navigation bar includes the Bibliometrix logo, a menu icon, and user information (1.42M, ?). The left sidebar contains navigation links: biblioshiny, Info, SEARCH, Data, APPRAISAL, Filters, PRISMA Diagram, ANALYSIS, Overview, Main Information, Annual Scientific Production, Average Citations per Year, Life Cycle, Three-Field Plot, and Sources.

The main content area is divided into several sections:

- Filter List**: Info & References
- Documents**: 167 of 193
- Sources**: 134 of 149
- Authors**: 675 of 743
- Buttons**: Apply, Reset, Data

The main content area is divided into four panels:

- 1. General**
 - Document Type**: ARTICLE, ARTICLE; EARLY ACCESS, CONFERENCE PAPER, PROCEEDINGS PAPER
 - Language**: ENGLISH
 - Publication Year**: 2013 to 2026
- 2. (J) Journal**
 - Upload a List of Journals**: Browse... No file selected
 - Upload a Journal Ranking List**: Browse... No file selected
 - Source by Bradford Law Zones**: All Sources
- 3. (AU) Author's Country**
 - Region**: Africa, Asia, Europe, North America, South America, Seven Seas, Oceania, Unknown
 - Country**: Search... (List of countries: AUSTRALIA, AUSTRIA, BANGLADESH, BRAZIL, CANADA, CHINA)
- 4. (DOC) Documents**
 - Total Citations**: 0 to 489
 - Total Citations per Year**: 0 to 62

Final takeaway for Week 2 第二周总结

SEARCH

Data <

APPRAISAL

Filters

PRISMA Diagram

ANALYSIS

Overview <

- Main Information
- Annual Scientific Production
- Average Citations per Year
- Life Cycle
- Three-Field Plot

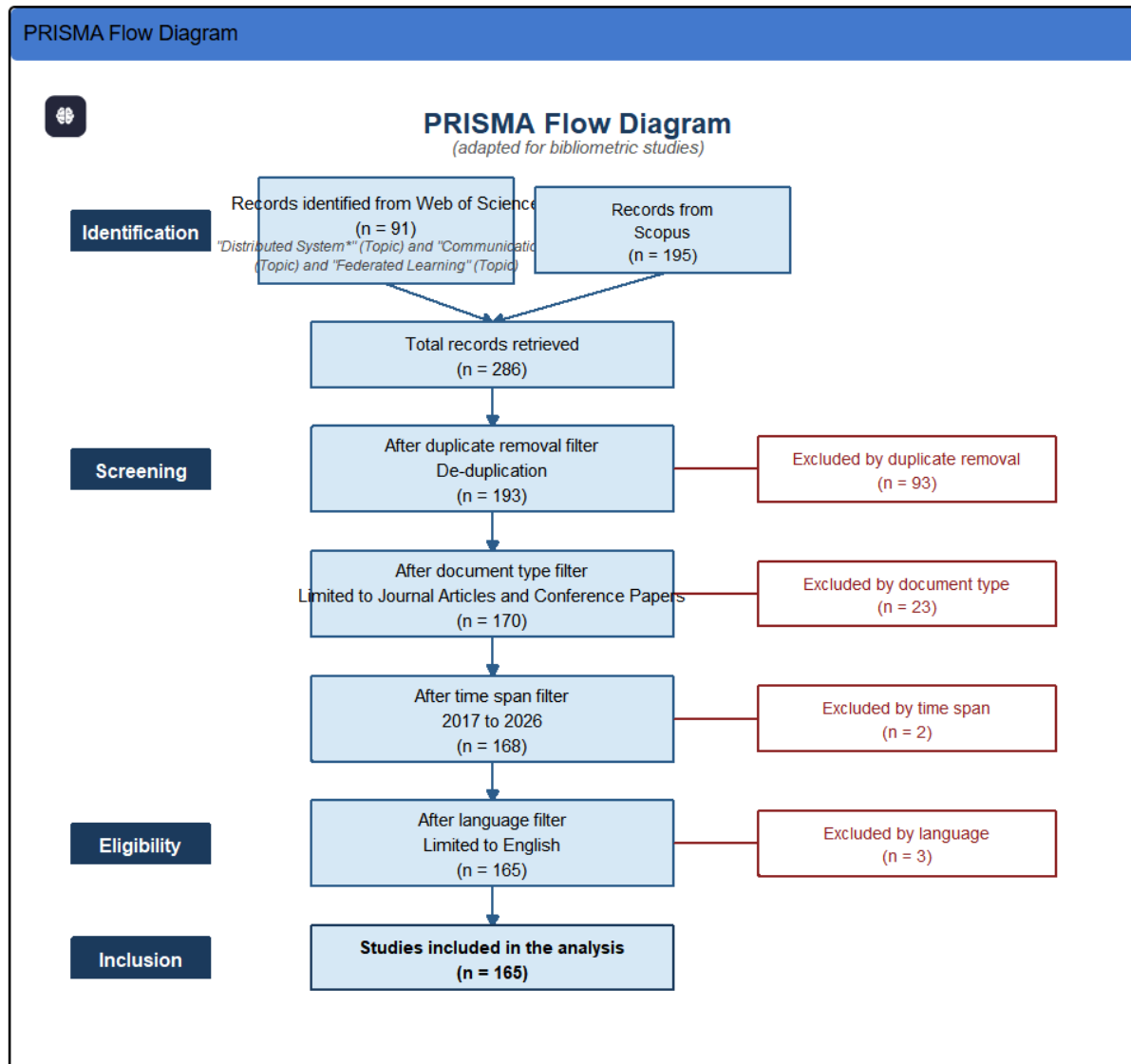
Sources <

Authors <

Documents <

SYNTHESIS

- Conceptual Structure <
- Social Structure <



Identification

Database Name

Web of Science

Search Query / Topic

"Distributed System*" (Topic) and "Communication*" (Topic)

Total Records Retrieved

91

Additional Source (optional)

Other Source Description

Scopus

Records from Other Source

195

Screening & Eligibility Steps

Add filtering steps in the order they were applied. For each step, select the criterion and enter the number of records remaining. The phase (Screening/Eligibility) and excluded counts are computed automatically.

Step 1

Filter Criterion

Duplicate removal

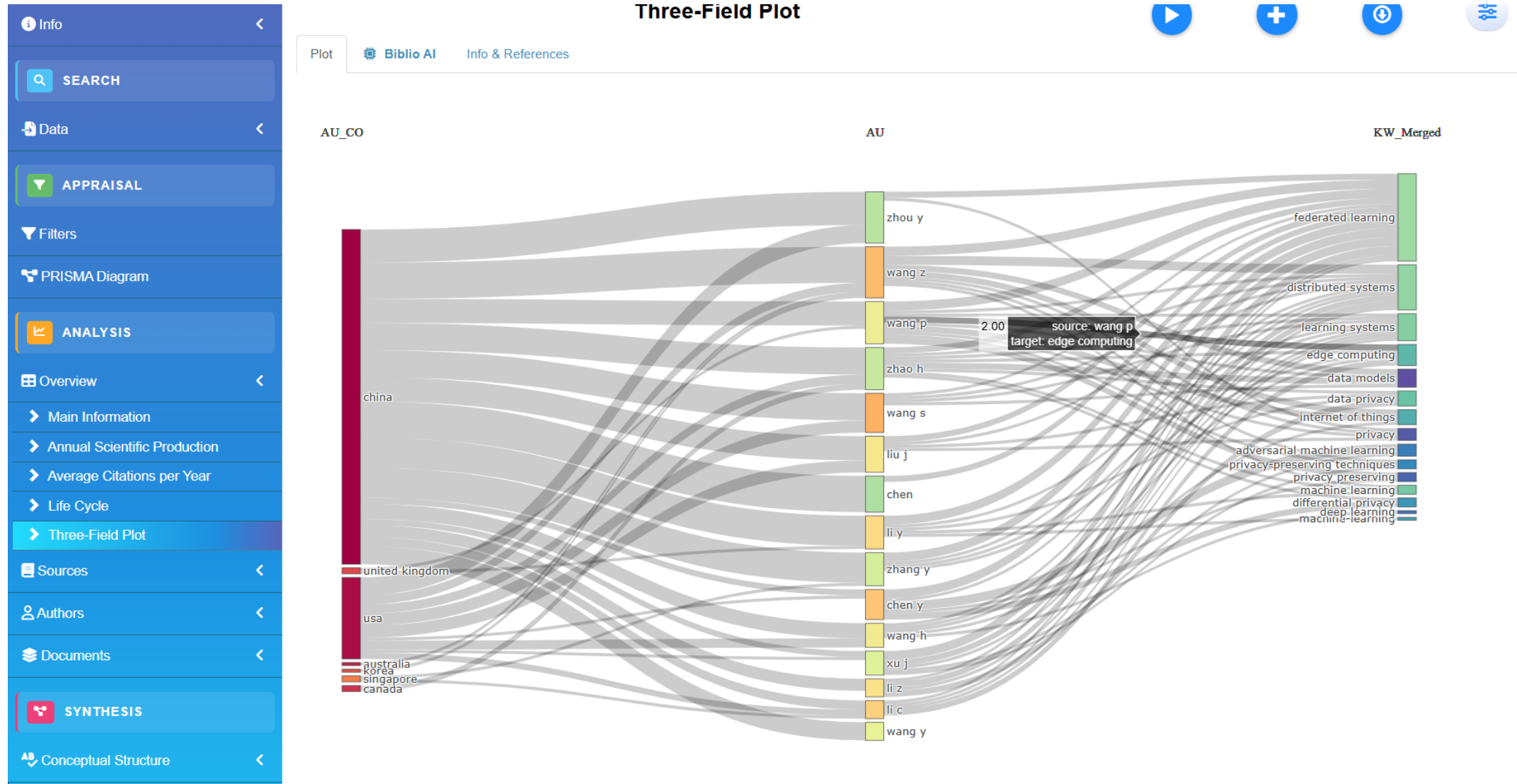
Filter Values (optional)

De-duplication

Week 5: Communication and Naming in Distributed Systems

第5周：分布式系统中的通信与命名

Final takeaway for Week 2 第二周总结



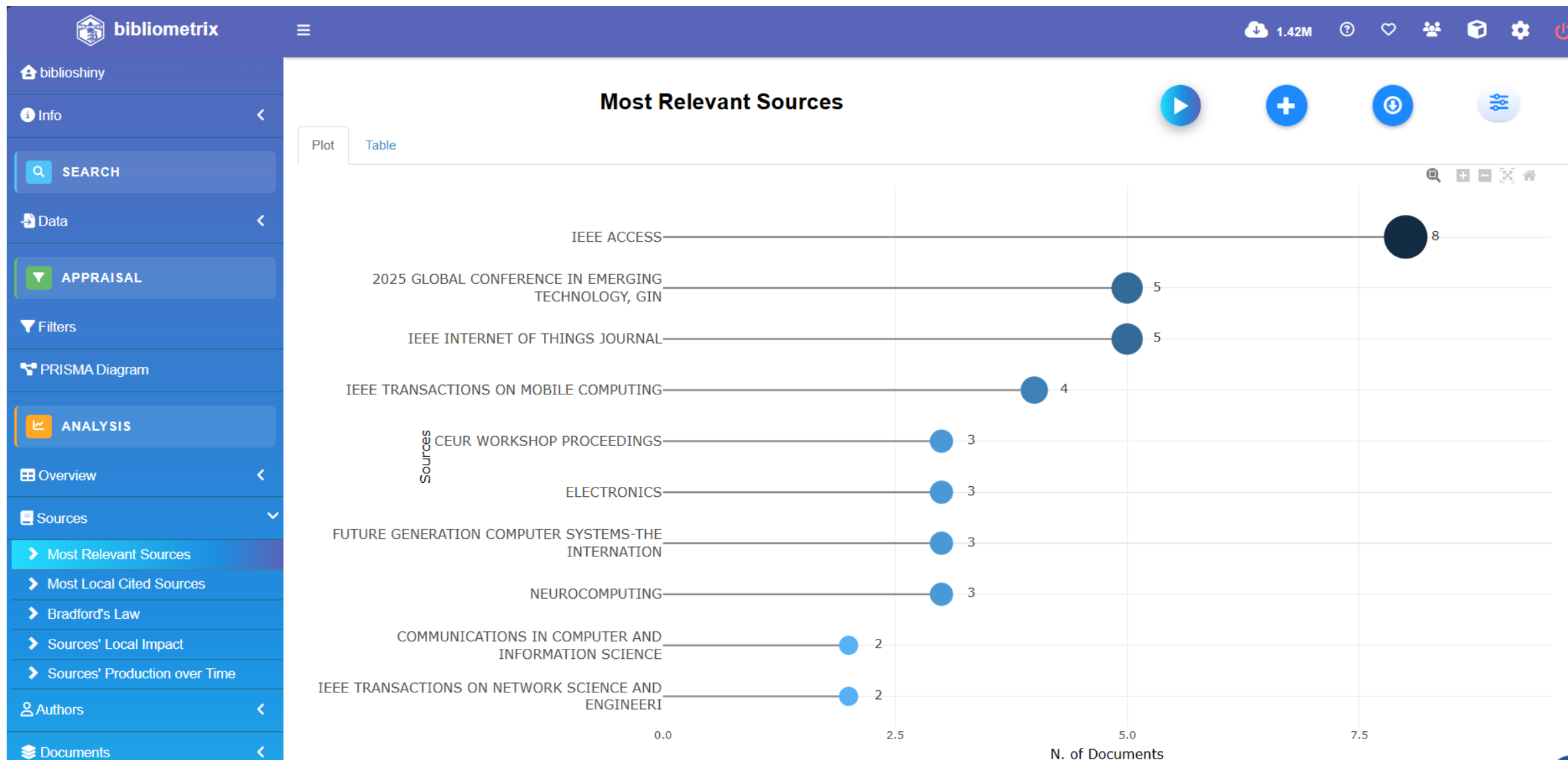
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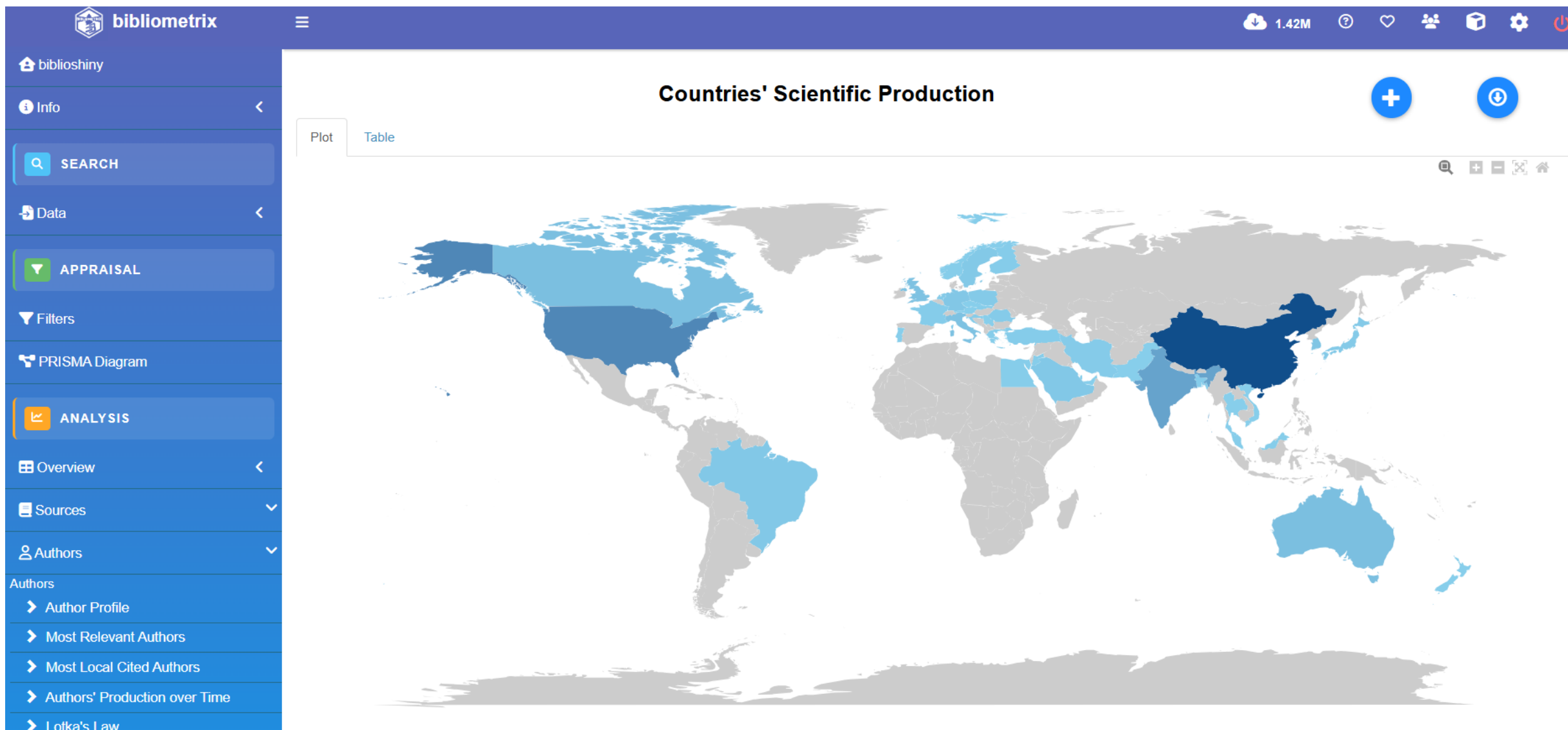
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第5周：分布式系统中的通信与命名

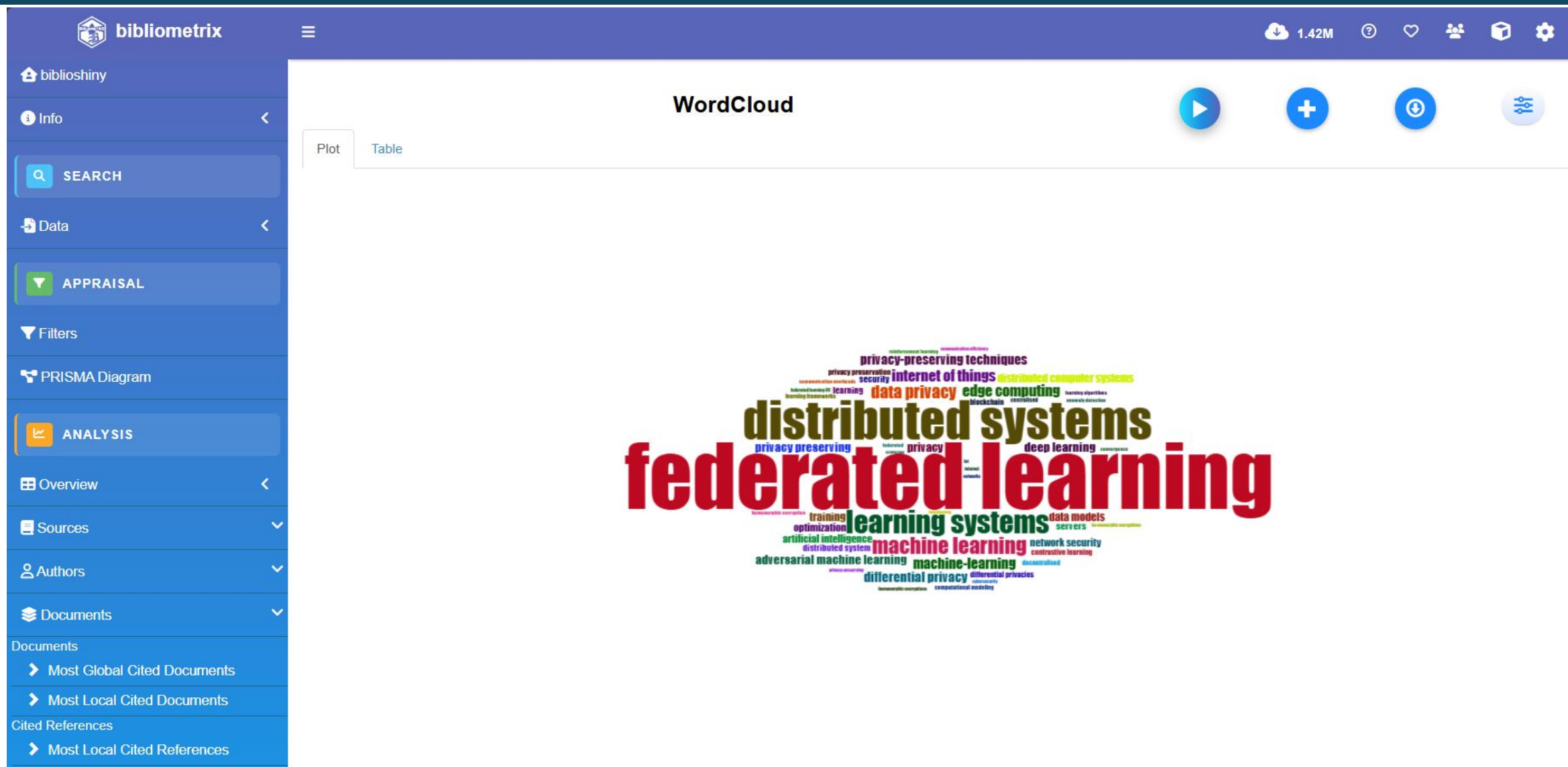
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Final takeaway for Week 2 第二周总结

The screenshot shows the bibliometrix web application interface. The left sidebar contains navigation links: biblioshiny, Info, SEARCH, Data, APPRAISAL, Filters, PRISMA Diagram, ANALYSIS, Overview, Sources, Authors, and a sub-menu for Authors (Author Profile, Most Relevant Authors, Most Local Cited Authors, Authors' Production over Time). The main content area is titled 'Countries' Scientific Production' and displays a table of scientific production by country. The table has two columns: 'Country' and 'Freq'. The data is sorted by frequency in descending order. The first 10 rows are shown, with a total of 39 rows. The countries listed are CHINA, USA, INDIA, AUSTRALIA, CANADA, ITALY, UK, SOUTH KOREA, GERMANY, and BRAZIL. The frequencies are 136, 73, 44, 16, 16, 13, 13, 10, 8, and 7 respectively. The interface also includes a search bar, a filter input, and a pagination control at the bottom right.

Country	Freq
CHINA	136
USA	73
INDIA	44
AUSTRALIA	16
CANADA	16
ITALY	13
UK	13
SOUTH KOREA	10
GERMANY	8
BRAZIL	7

Final takeaway for Week 2 第二周总结



Week 5: Communication and Naming in Distributed Systems

第5周：分布式系统中的通信与命名

北京工业大学 分布式系统的原理

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